Mutual Fund Trading Style and Bond Market Fragility *

Amber Anand
Syracuse University
amanand@syr.edu

Chotibhak Jotikasthira,
Southern Methodist University
cjotikasthira@mail.smu.edu

Kumar Venkataraman
Southern Methodist University
kumar@mail.cox.smu.edu

Abstract

We explore the link between mutual funds and fragility risk in the corporate bond market. We classify a fund's trading style based on its responses to signals of large dealer inventories. Trading style is persistent and the majority of funds demand liquidity. Notably, a subset of funds earn positive alpha by intentionally supplying liquidity during periods of sustained customer selling (with transitory price effects). Liquidity supplying funds maintain their relative trading style when facing large outflows and elevated market stress, thus alleviating fragility risk. Our results add nuance to existing evidence that mutual funds pose a threat to market stability.

Keywords: Corporate bonds; dealers; bond mutual funds; fragility; performance.
This Draft: May 7, 2020

<u>*Review of Financial Studies*</u>*, forthcoming.*

---

* A previous version of this paper circulated under the title "Do buy-side institutions supply liquidity in bond markets? Evidence from mutual funds."

We thank Hank Bessembinder, Yong Chen, Zhi Da, Jack Bao, Jaewon Choi, Paul Calluzzo, Peter Feldhutter, Itay Goldstein (the Editor), Stacey Jacobsen, Ryan Israelsen, Eric Kelley, Amar Kuchinad, David Krein, Bill Maxwell, Mike Piwowar, Andy Puckett, Alex Reyfman, Tavy Ronen, Marco Rossi, Mehrdad Samadi, Norman Schurhoff, Pradeep Yadav and two anonymous referees for their comments. We are also grateful for comments received during 2018 Chicago Financial Institutions conference, 2018 Bank of Canada-Laurier Market Structure conference, 2019 European Financial Association conference, 2017 Emerging Markets Finance conference, 2017 Financial Economics and Accounting conference, 2018 Financial Intermediation Research Society Conference, 2018 Midwest Finance Association conference, FINRA/Columbia University Market Structure and Capital Formation conference, ESSEC Business School, Babson College, Rochester Institute of Technology, Texas A&M University, University of Illinois Urbana Champaign, University of Oklahoma, Universite Paris-Dauphine, University of Tennessee and Southern Methodist University for their comments. The authors thank the Finance Industry Regulatory Authority (FINRA), and in particular, Alie Diagne, Ola Persson, Jonathan Sokobin and Chris Stone for provision of data. Anand and Venkataraman are affiliated as visiting economists with the office of chief economist at FINRA.

Mutual Fund Trading Style and Bond Market Fragility

Abstract

We explore the link between mutual funds and fragility risk in the corporate bond market. We classify a fund's trading style based on its responses to signals of large dealer inventories. Trading style is persistent and the majority of funds demand liquidity. Notably, a subset of funds earn positive alpha by intentionally supplying liquidity during periods of sustained customer selling (with transitory price effects). Liquidity supplying funds maintain their relative trading style when facing large outflows and elevated market stress, thus alleviating fragility risk. Our results add nuance to existing evidence that mutual funds pose a threat to market stability.

I. Introduction

The financial stability of the fixed income markets, and the potentially destabilizing role of institutional investors, have received considerable attention in recent years. Between 2008 and 2018, the corporate bond market nearly doubled in size from $5.1 trillion to $9.2 trillion, alongside significant increases in assets under management by mutual funds and exchange traded funds (ETFs) that invest in corporate bonds, as shown in Figure 1.[1] The recent low-yield environment has pushed bond funds to 'reach-for-yield' by holding less liquid assets (IMF (2019)). Unlike equity funds, poor performance leads to large investor outflows from bond funds, raising concerns about price destabilizing effects of funds selling relatively illiquid assets to meet redemption needs (Goldstein, Jiang and Ng (2017)). Further, institutions tend to buy or sell corporate bonds in a correlated manner, and large trade imbalances attributable to institutional herding can lead to price distortions, particularly on the downside (Cai, Han, Li and Li (2019)).

The stability risks posed by liquidity demand from funds are heightened in light of recent evidence that post-crisis banking-related regulations, such as Volcker Rule, have reduced dealer capital for market making in fixed income securities (Bessembinder, Jacobsen, Maxwell, and Venkataraman (2018), Bao, O'Hara, and Zhou (2018), Dick-Nielsen and Rossi (2019), and Schultz (2017)). Regulators are concerned that incidents of institutional selling coupled with reductions in the supply of dealer capital can potentially amplify the fragility risk of the bond market.[2]

Could institutional investors be part of the potential solution to fragility risk? In contrast to the regulatory view that institutions, particularly bond mutual funds (Manconi, Massa, and Yasuda (2012)),

[1] Statistics on outstanding corporate bonds are obtained from SIFMA 2019 Factbook (http://www.sifma.org/research). U.S. equity issuance in 2018 amounted to $220 billion while corporate bond issuance amounted to $1.3 trillion. Investment Company Institute (ICI) data show that bond mutual funds and bond ETFs share of the corporate bond market has more than doubled from 7.3% in 2006 to 17.9% in 2016.
https://www.ici.org/viewpoints/view_17_corp_bond_etf

[2] Among the stability risks identified by the Financial Stability Oversight Council (2016), fund liquidity and redemptions are listed as the most significant. The U.S. Securities and Exchange Commission (SEC) has issued guidelines for bond fund to "*assess funds' liquidity, and the ability to meet potential redemptions, during both normal and stressed environments, including assessing their source of liquidity*". In January 2018, the SEC established a sub-committee within Fixed Income Market Structure Advisory Committee (FIMSAC) to assess the impact of bond funds redemptions on bond market fragility. https://www.sec.gov/divisions/investment/guidance/im-guidance-2014-1.pdf.

pose risks to financial stability, market participants appear to hold the view that institutions play a balanced role in fixed income markets. For example, Greenwich Associates reports that 78% of the credit investors participating in their recent survey identify buy-side institutions as an important source of liquidity in the corporate bond market.[3]

To better understand the linkage between bond institutions and fragility risk, we need more research on the role that institutions play in destabilizing or improving liquidity conditions in bond markets, particularly during periods of market stress. A surge in simultaneous customer buying or selling causes dealers to accumulate large positions, and given dealers' limited market-making capacity, longer-term investors can play a role in absorbing market imbalances. Institutions do not necessarily incur inventory costs on positions that meet their long-term investment objectives, and this feature uniquely positions them to serve this role.

We develop a methodology to classify the trading style of bond institutions, leveraging unique features of bond markets and available data. We focus on corporate bond mutual funds, which are the fastest growing category of institutional investors in the corporate bond market. They trade more actively than others and are likely to benefit more from being sensitive to liquidity conditions. We study how funds respond to signals of market imbalances; whether some funds strategically choose to supply liquidity and profit from doing so; and whether fund attributes explain trading style. To understand the implications for fragility risk, we study whether trading style is altered, and whether liquidity supplying and demanding funds differentially impact the bond markets during periods of extreme flows and elevated market stress.

This study fills a gap in the bond market literature on the supply and demand of liquidity services by institutions. Existing studies on institutional trading have focused on equity markets (e.g., Da, Gao and Jagannathan (2011), Anand, Irvine, Puckett and Venkataraman (2012, 2013), among others); however, bond markets differ from equity markets in several important ways. First, as shown in Goldstein et al. (2017), the potential for instability posed by mutual funds is a more pressing concern for bond markets than

[3] "Innovations ease corporate bond trading", Greenwich Associates, Quarter 2, 2017.

equity markets. For equity funds, the flow-to-performance relation has a convex shape, implying that investors do not rush to withdraw funds after poor performance, and further, the relation is essentially flat for the aggregate fund sector. On the other hand, for bond funds, the flow-to-performance relation is concave, both for individual funds and the aggregate fund sector, indicating that poor performance leads to large flows out of the bond sector.

Second, in contrast to equities which trade frequently, the majority of corporate bonds are illiquid, implying that order flow imbalances attributable to herding or flows have a greater potential to destabilize bond markets. Cai et al. (2019) show that price impact of institutional herding in corporate bonds, especially on the sell side, is substantially higher than that observed in equities. Third, unlike equities which trade on exchanges, corporate bonds trade in over-the-counter (OTC) markets, making it difficult for institutions to gather pre-trade information, and to directly supply liquidity by posting limit orders.[4] These differences in market structure between equity and bond markets suggest that trading strategies, such as liquidity provision, that work in equity markets are challenging to implement in bond markets. Thus, whether buy-side liquidity provision exists in bond markets and how it interacts with fragility remain open questions.

The framework of liquidity supply that we propose hinges on the notion of trading against sustained customer imbalances. We measure imbalances using the enhanced version of the TRACE database, which, in addition to reporting all U.S. corporate bond transactions reflected in public TRACE, also includes uncapped trade sizes and masked dealer identifiers. Unlike the equities TAQ database that does not identify market makers, the TRACE data capture the entire history of dealers' trades with customers, allowing us to measure inventory positions of market intermediaries in corporate bonds. We classify a scenario with sustained customer selling (buying) activity as a positive (negative) inventory cycle to reflect the positive (negative) inventory positions of dealers. We identify the beginning and ending dates of an inventory cycle in a bond when cumulative dealer inventory crosses zero. Cumulative inventory is the signed, aggregate dollar inventory in a bond based on all dealers' trades with customers.

[4] Recent industry initiatives aim to bring greater pre-trade transparency to bond markets. See, for example, "Barclays, FactSet Offer New Tool to Navigate Opaque Bond Market", *Wall Street Journal*, January 15, 2020.

Over the 2002-2017 sample period, the inventory cycles that we identify are long lasting, with the average cycle length of over 70 calendar days and peak inventory over $20 million.[5] Cai et al. (2019) find that a surge in institutional selling is triggered by credit rating downgrades and poor past bond performance. Both events are indeed associated with a higher likelihood of positive cycles, as shown in Figure 2. Cai et al. (2019) also document a striking pattern of buy-sell asymmetry where buy herding results in a permanent increase in bond prices while sell herding results in a transitory but significant decline in bond prices. Figure 2 shows that bonds in negative cycles are associated with permanent price adjustments (i.e., price discovery) while price changes of bonds in positive cycles are associated with reversals (i.e., transitory price pressure).

We classify trading style based on a fund's response to signals of market imbalances as reflected in dealer inventory cycles. Fund transactions are identified by the change in bond holdings over a fund's reporting period (Da et al. (2011)), and classified as liquidity supplying, demanding, or unclassified by overlaying the trade on a bond's inventory cycle – liquidity supplying trades absorb dealer positions while liquidity demanding trades strain dealer positions.[6] Our methodology differs from approaches in the equity market literature that classify trading style based on the relation between fund trades and the daily stock return (Anand et al. (2013)) or the daily TAQ trade imbalance (Da et al. (2011)). Given bond illiquidity, the slow-moving inventory cycles present an accurate picture of liquidity conditions in a bond. Further, unlike returns and trade imbalances that reverse sign within a cycle, the cycle-based classification is not sensitive to whether a fund-reporting period overlaps with the build-up or unwind phases of the cycle. Our approach is subject to data limitations such as inability to classify transactions in securities that are not TRACE-eligible, and similar to other approaches, the lack of information on the precise timing of fund transactions.

For each fund-reporting period, we aggregate the dollar amount of liquidity supplied and demanded across all corporate bond transactions and obtain a fund's composite trading style, *LS_score*. Higher values

[5] In equities, the holding period of active market makers (i.e., high-frequency traders) is very short (in seconds) while in corporate bonds, Schultz (2017) estimates that half-life of individual dealer inventory positions is four to five weeks.
[6] A transaction is liquidity supplying (demanding) if the fund buys (sells) a bond when dealers face sustained customer selling or sells (buys) a bond when dealers face sustained customer buying.

of *LS_score* indicate a trading style that helps alleviate large dealer positions. Based on the *LS_score*, we classify the funds in the highest quintile as liquidity suppliers (LS), and those in the lowest quintile as liquidity demanders (LD).

Our main results are as follows. First, consistent with prior studies, the typical mutual fund exhibits a trading style that strains large dealer positions. This result suggests that mutual funds, on average, contribute to the market imbalances observed from all customer trades that we use to build inventory cycles. Trading style varies in the cross-section of funds and past style is informative about future style. We identify a subset of funds (LS) with a trading style that absorbs and/or avoids straining dealer positions.

Second, a liquidity supplying trading style contributes positively to fund alpha after controlling for known determinants of fund performance including lagged fund alpha, a proxy for unobservable fund manager skills. LS funds outperform LD funds by two basis points per month. For context, the average monthly gross alpha of our sample funds is 3.40 basis points.[7] Notably, when we decompose trading style into that attributable to positive and negative cycles, the contribution to alpha is associated with liquidity supply during positive cycles. As shown in Figure 2, positive cycles are associated with transitory movements in bond prices. These results support the interpretation that the returns attributable to trading style reflect compensation for supplying liquidity, which are distinct from the liquidity premium earned for holding illiquid bonds. Stated differently, the returns reflect *how* institutions build and unwind positions, rather than *which bonds* they hold in the portfolio.

Third, the decision to supply liquidity is a strategic choice and does not arise simply from funds' actions to accommodate time-varying flows that coincide with incidents of positive cycles. Among factors that facilitate a liquidity supplying trading style are stable investor base (Giannetti and Kahraman (2018), Jylha, Rinne and Suominen (2014), and Franzoni and Plazzi (2015)), cash holdings and better information about market conditions. Specifically, funds affiliated with a broker-dealer, younger and smaller funds ((Da et al. (2011) and Cremers and Petajisto (2009)), and funds with larger fraction of institutional share classes,

[7] The average gross alpha is similar in magnitude to that reported by Cici and Gibson (2012).

lower flow-performance sensitivity, larger cash holdings, and larger fund flows are more likely to supply liquidity. Past trading style remains predictive of future fund behavior after accounting for these factors, suggesting that trading style is a persistent fund attribute.

Finally, and perhaps most importantly, while prior work concludes that mutual funds pose a threat to market stability, our study is the first, as far as we know, to show that some funds could potentially help alleviate the fragility risk of the bond markets.[8] Two important factors that amplify the risks to financial stability are institutional selling that is driven by large fund outflows and the decline in dealer capital during episodes of market stress. We study the trading behavior of bond funds under these stress scenarios.

We examine individual funds' selling behavior in the face of extreme outflows, and find that LS funds maintain their trading style, relative to others. Specifically, LS funds are significantly less likely to sell bonds in positive cycles and more likely to sell bonds in negative cycles. Following Edmans, Goldstein and Jiang (2012), we build a naïve flow induced selling measure where a fund facing extreme outflows sells bonds in proportion of its holdings. Flow induced selling by LD funds significantly increases a bond's probability of being in a positive cycle, but flow induced selling by LS funds does not. Similarly, flow induced selling of LD funds is associated with a price impact that is over 2.5 times larger than the price impact associated with flow induced selling of LS funds. We also examine funds' trading behavior when dealers face capacity constraints, identified as periods with high proportions of corporate bonds in positive cycles and high values of Volatility Index (VIX). During these periods, LS funds help alleviate dealer stress both in their buying (i.e., buy more bonds in positive cycles than those in negative cycles) and selling (i.e., sell fewer bonds in positive cycles than those in negative cycles) activities.

Collectively, our study offers a nuanced view of the potential threats to bond market stability posed by mutual funds and highlights institutions as a potential liquidity source in bond markets. As institutions represent a predominant share of the bond investor base, the economic importance of buy-side liquidity

[8] In equity markets, Jylha et al. (2014) and Franzoni and Plazzi (2015) find that hedge funds switch from supplying liquidity in normal times to consuming liquidity in times of stress, while Anand et al. (2013) find that liquidity supplying institutions tilt away from participating in illiquid securities during the financial crisis. Our results show notable differences in the behavior of liquidity supplying institutions in bond markets.

provision cannot be overstated. To tap into, and further encourage, this channel of liquidity supply, fixed income platforms need to offer trading protocols that allow institutions to compete with traditional dealers for liquidity provision. As noted by former SEC Commissioner Michael S. Piwowar, while bond trading platforms hold much promise, the majority of venues, even in recent years, restrict the customer's request-for-quotation (RFQ) messages to participating bond dealers.[9] Our study highlights improvements in market design that can enhance the financial stability of bond markets.

II. Related work and hypotheses development

Since July 2002, SEC-registered broker-dealers are required to report all transactions that they facilitate in corporate bonds, as principal or agent, to FINRA's TRACE system. With the availability of TRACE corporate bond data, academic studies have documented several notable liquidity patterns: (a) Customer trading costs in corporate bonds are large relative to those observed in equity markets (e.g., Schultz (2001), Ederington, Guan and Yadav (2015), Harris (2016)),[10] (b) Introduction of TRACE transactions reporting system has lowered trading costs (Bessembinder, Maxwell, and Venkataraman (2006), Edwards, Harris, and Piwowar (2007), Goldstein, Hotchkiss, and Sirri (2007)), (c) Corporate bond liquidity declined during the 2007-09 financial crisis and contributed to higher bond yields (Friewald, Jankowitsch and Subrahmanyam (2012)), (d) Electronic bond platforms have gained market share for large issues, actively traded bonds and smaller transaction sizes (Hendershott and Madhavan (2015)), and (e) Post-crisis banking regulation led to a decline in capital commitment by bank-affiliated bond dealers (Bessembinder et al. (2018), Bao et al. (2018)). Bessembinder, Spatt, and Venkataraman (2020) present a survey of the academic evidence on the microstructure of the fixed income markets.

Regulatory concerns that bond mutual funds pose a greater threat than equity mutual funds to financial stability are attributable, at least in part, to bond market illiquidity. Illiquidity creates a mismatch

[9] https://www.sec.gov/news/speech/piwowar-remarks-finra-2016-fixed-income-conference.html
[10] For corporate bonds, Harris (2016) reports average effective half spreads of 65 basis points for customer trades (Table 14) while for US equities, Chordia, Roll and Subrahmanyam (2011) report effective half spreads lower than five basis points before the financial crisis (Figure 4). Anand et al. (2013) report that effective spreads return to pre-crisis levels by 2010.

between the daily liquidity offered by funds on investor redemptions and their ability to liquidate holdings at a low cost. Chen, Goldstein and Jiang (2010) argue that the policy to redeem funds at NAV transfers liquidation costs from redeeming investors to other investors who keep their money in the fund. Investors in funds that hold illiquid assets are particularly sensitive to the wealth transfer, which leads to a first-mover advantage and amplification of exits for these funds following bad performance.

The equity mutual fund literature shows that the flow-to-performance relation has a convex shape, implying large fund inflows in response to good past performance but relatively little sensitivity in response to bad past performance (Chevalier and Ellison (1997) and Sirri and Tufano (1998)). For corporate bond funds, Goldstein et al. (2017) find a concave flow-performance relation, wherein outflows are more sensitive to bad past performance than inflows are to past good performance, attributable in part to the first-mover advantage. In addition, Cai et al. (2019) show that institutional herding in corporate bonds is stronger than that observed in equities, and due to bond illiquidity, herding leads to larger price effects in corporate bonds, particularly on the sell side.[11]

Bond illiquidity creates an opportunity to earn alpha for fund managers with the skill to incorporate liquidity conditions into portfolio decisions. Microstructure theory predicts that liquidity provision is profitable around price pressure events with transitory effects and not information events with permanent price effects (Kraus and Stoll (1972), and Grossman and Miller (1988)). In the case of *naïve* liquidity supply, the fund supplies liquidity in response to imbalance signals but does not differentiate between price pressure and information events. In the case of *sophisticated* liquidity supply, the fund responds only to price dislocations that are transitory, both earning the price impact by trading against imbalance, and limiting the price impact by avoiding trading with imbalance. In the context of our research design, given the evidence in Figure 2, *sophisticated* liquidity supply will be observed around price pressure (positive cycles) and not price discovery (negative cycles) events, and the former will contribute to fund performance.

To the extent that a fund buys and sells bonds in response to investor flows, trading against market

[11] Explanations for institutional herding include bond performance (Cai et al. (2019)), flows (Goldstein et al. (2017)), index rebalancing (Dick-Nielsen and Rossi (2019)) and credit rating events (Ellul, Jotikasthira and Lundblad (2011)).

imbalances may not be an *intentional* decision to supply liquidity but rather an *accidental* outcome (Da et al. (2011)). Suppose a mutual fund experiences large inflow when many bonds are in positive cycles. The buying activities of the fund in response to inflows will absorb dealer positions even though liquidity supply is not the motivation behind the trades. If liquidity supply is an *intentional* decision, trading style should emerge as a fund-specific attribute that persists after controlling for the timing of investor flows that coincides with sustained market imbalances. These discussions lead to the following hypotheses:

Hypothesis H1 – *Trading style is a fund-specific attribute and varies in the cross-section of funds. Trading style captures a decision to supply liquidity that is intentional and sophisticated.*

Hypothesis H2 – *Liquidity supply contributes positively to fund alpha, and the relation is driven by fund's trading in bonds that experience positive cycles.*

The existence of positive alpha to a liquidity provision strategy points to either implementation costs or institutional barriers that limit wider adoption of the strategy. The OTC structure of bond markets makes it difficult for funds to acquire information on market conditions and to supply liquidity using limit orders. A relationship with bond dealers (e.g., an affiliated dealer desk) can offer fund managers with information flow that is difficult for the peers to replicate. A stable investor base that is less sensitive to past performance can enable funds to supply liquidity, as the fund is less likely to be forced into costly trading in response to investor flows, both as a result of liquidity shocks (Cella, Ellul, and Giannetti (2013), Giannetti and Kahraman (2018), and Manconi et al. (2012)) and performance chasing (Huang, Wei, and Yan (2012)). Further, funds can reduce impact of fund redemptions by holding cash (Chernenko and Sunderam (2018)), or more generally liquid assets, and restricting redemptions, which might enhance their ability to supply liquidity.

Hypothesis H3 – *Funds with stable investor base, liquid holdings, and close relationships with dealers are more likely to supply liquidity.*

Two important factors that are often cited as the drivers of fragility risk in bond markets are large fund outflows (and their feedback effects) and the sharp decline in dealer capital when markets are stressed. The fire-sale literature shows that equity funds, experiencing extreme outflows, sell almost proportionally across holdings (Coval and Stafford (2007)) while bond funds trade in a more nuanced manner, dynamically trading off price impact against liquidity preservation (Jiang, Li, and Wang (2019), and Choi, Hoseinzade, Shin, and Tehranian (2019)). To better understand fragility risk posed by bond funds, it is important to study fund behavior during periods of stress, and to examine whether liquidity supplying funds maintain their trading style when conditions are challenging. This leads to the following testable hypothesis:

> <u>Hypothesis H4</u> – *Liquidity supplying funds maintain their relative trading style under difficult conditions. Funds with different trading styles differentially contribute to bond market fragility.*

III. Sample selection and inventory cycles

A. Data and sample

The study's primary data sources are as follows. We obtain data on corporate bond transactions from FINRA's enhanced TRACE database, data on bond characteristics from Mergent's Fixed Income Securities Database (FISD), data on bond mutual fund holdings from Morningstar, and the *VIX* index from the Chicago Board Options Exchange (CBOE).

The public version of the TRACE data includes information on bond CUSIP, the date and time of execution, the transaction price and volume (in dollars of par), and whether the trade represents a sale or purchase of bonds by a dealer to a (non-dealer) customer, or a trade between two dealers, and for customer trades, whether the customer is a buyer or a seller. The enhanced TRACE data made available to academics by FINRA include information on disseminated and non-disseminated historical transactions, including those in non-registered 144A bonds; unmasked trade sizes that are capped in the public version for large transactions; and masked identification numbers for individual dealers participating in a transaction.

We obtain information on bond characteristics such as issue size, credit rating, and age from FISD database. Between July 2002 and March 2018, the TRACE database includes over 152,757 unique cusips. The majority of cusips pertain to instruments other than corporate bonds. Following Bessembinder et al. (2018), we identify 32,842 corporate bonds in FISD database that are classified as non-puttable U.S.

Corporate Debentures and U.S. Corporate Bank Notes (bond type-CDEB or USBN).[12] For these bonds, we select all transactions data between July 2002 and March 2018, and impose the following screens: (a) exclude bonds with less than five trades in the sample period, (b) exclude bonds with a reported trade size that exceeds the bond's offer size, (c) exclude trades that are reported after the bond's amount outstanding is reported by FISD as zero, and (d) exclude transactions that are flagged as primary market transactions. With these filters imposed, the sample comprises 98.7 million transactions in 32,006 distinct cusips.

From Morningstar, we obtain data on fund holdings and monthly (inferred) flows and returns for taxable bond mutual funds between July 2002 and June 2017. We focus on Morningstar's defined categories for which corporate bonds form a material part of portfolio holdings (average proportion of 30% or greater). These include Corporate Bond, High-Yield Bond, Multi-sector Bond, Nontraditional Bond, Bank Loan, Preferred Stock, Short-Term Bond, Intermediate-Term Bond, and Long-Term Bond funds.[13]

We present descriptive statistics for the sample of bond funds in Table 1. While the SEC requires mutual funds to report holdings on a quarterly basis, some funds choose to report their holdings even more frequently. Approximately 72% of the fund reporting-periods in our sample are monthly. We do not filter funds based on reporting frequency; instead, we condition on the available frequency in measuring trading style. Panel A reports the statistics for 72,632 fund-reporting period observations in the sample. The snapshots of fund holdings show an average (median) of 463 (276) positions, representing total net assets (TNA) of approximately \$1.9 billion (\$0.38 billion). Figure 1 shows the number and aggregate TNA of funds in our sample over time. Consistent with Goldstein et al. (2017), bond funds' holdings increase over time, from about \$600 billion in 2002 to \$2.5 trillion in 2017.

[12] Stated differently, we exclude cusips that pertain to retail notes, foreign government bonds, U.S. agency debentures, asset backed securities, pay-in-kind bonds, medium term notes, convertible and preferred securities.
[13] We choose the 30% threshold to allow for a balanced sampling of investment grade and high yield bond funds. Investment grade bond funds invest more broadly across different types of bonds (e.g., Treasury, Agency, and MBS). At a higher threshold, our sample will skew towards high yield bond funds since the majority of high yield bonds tend to be corporate bonds. The 30% threshold is based on Morningstar's category average to avoid selecting individual funds based on their strategies. As robustness, we examine a sample of bond funds that hold at least 50% of portfolio in corporate bonds and find similar results. The average corporate bond holdings for this sample is 81%.

In addition to corporate bonds, bond mutual funds invest in other securities such as government bonds, international bonds, and structured products. The average (median) fund in the sample invests 50% (43%) of its portfolio in corporate bonds, of which corporate bonds included in our TRACE sample account for 40% (31%) of TNA. Sample funds hold an average of 9% of TNA in cash and cash equivalents (as defined by Morningstar), 14% in government bonds, and 26% in other securities. For the fund family, the average TNA is $8.9 billion with six funds per family in the sample. Consistent with Cici and Gibson (2012) and Goldstein et al. (2017), most bond funds are actively managed, with only 3.2% of funds identified as index funds. The median bond fund has no rear load fee, and institutional share classes account for 21% of TNA. The median monthly fund flow is 0.05% of TNA with the 25^{th} and 75^{th} percentile at -1.2% and 1.6%, respectively, indicating significant variation across funds and over time.

Following the approach used by Da et al. (2011) for equity markets, we infer the transactions of the bond fund by comparing a fund's holdings in a bond over consecutive reporting periods. Studies on institutional trading in US equities have relied on Plexus and Ancerno data on institutional transactions, but no similar data exist for bond transactions of mutual funds and pension funds.[14]

Table 1, Panel B, reports statistics on funds' turnover (annualized), broken down by frequency of reporting. In our sample, about 72% (i.e., 52,422 out of 72,632) of fund-reporting period observations are monthly and 26% are quarterly (i.e., 18,704 out of 72,632). We calculate the turnover as the change in holdings, including both increases and decreases, in a reporting period, excluding bonds' expiration, divided by the total holdings at beginning of the period. Funds with monthly fund reporting have average (median) annualized turnover of 225% (151%) while the comparable statistic for funds with quarterly reporting is 190% (141%). Thus, bond mutual funds trade frequently, which differentiates them from the typical buy-and-hold bond institutions (Massa, Yasuda, and Zhang (2013)). Corporate bonds included in the TRACE sample account for 37% of total fund position changes in a fund-reporting period.

[14] While Ancerno does have a bond file, the data are extremely sparse.

B. Inventory cycles

Microstructure theory predicts that dealers manage inventory risk by lowering the ask price when inventory position is larger than desired and increasing the bid price when inventory position is smaller than desired (e.g., Stoll (1978), and Amihud and Mendelson (1980)). The "quote shading" behavior attracts the desired buyers or sellers and causes dealer inventory positions to revert to desired levels.[15] Based on dealer transactions with customers, we construct inventory cycles that represent episodes when dealers assume large inventory positions. A positive cycle is an episode in a bond with sustained customer selling, which leads to large positive inventory of dealers. Similarly, sustained customer buying leads to large negative inventories, which we term as a negative cycle.[16] Inventory cycles are not observed when customer buying and selling activity is balanced, when dealers act as brokers and match one customer with another (Harris (2016) and Choi and Huh (2019)), or when dealers quickly offset a customer trade with another customer. Interdealer trading serves an important channel for risk sharing among bond dealers (Schultz (2017)); however, interdealer trades do not alleviate aggregate customer imbalances, and are not considered in constructing cycles.[17]

Using TRACE transactions data, we calculate the (signed) aggregate dealer inventory in a bond by accumulating dealers' trades with customers from the start date of the cycle. We assume that the cycle starts when the cumulative inventory crosses zero, and ends when it crosses zero again from the opposite direction.[18] If the cycle remains ongoing and becomes longer than three months (63 trading days), then inventory is the (signed) cumulative customer imbalance over the rolling three months, which helps reduce the cycle's sensitivity to reporting errors that may otherwise compound infinitely. We select a rolling three-

[15] Empirical evidence on inventory management are presented by Hansch, Naik, and Viswanathan (1998) for London Stock Exchange dealers and Panayides (2007) for New York Stock Exchange specialists.
[16] Negative cycles might reflect that some dealers take a short position. Asquith, Au, Covert and Pathak (2013) show that borrowing costs for corporate bonds are comparable to those for stocks and have fallen over time.
[17] We consider all customer trades including those trades that are reported to TRACE as "Agency" trades. When a dealer acts as an agent, the dealer reports two legs of the facilitated trade as separate transactions on TRACE. When both legs involve customers, the net impact on the aggregate dealer positions is zero. When one leg involves a customer and the other leg involves a dealer, we include the customer leg but not the inter-dealer leg.
[18] In doing so, we implicitly assume that the desired (unobservable) dealer inventory is zero, a common assumption in microstructure research. Zero inventory is intuitively appealing because inventory requires capital commitment and exposes the dealer to volatile prices.

month period to allow for the slow build-up and unwinding of inventory in an illiquid market. Appendix 1 provides the details of our methodology. We require inventory to be material by imposing a minimum peak inventory of $10 million and a minimum inventory cycle length of 5 calendar days.

Table 2 summarizes the 207,871 inventory cycles identified by our methodology. There are 117,825 positive cycles, representing dealer buying activity to accommodate sustained customer selling, and 90,046 negative cycles, representing dealer selling activity to accommodate sustained customer buying. Inventory cycles are long lasting, with average (median) cycle lengths of 69 (64) calendar days for positive cycles, and 71 (65) calendar days for negative cycles.[19] The length of the loading and unloading phase of the cycle is similar indicating that peak inventory is observed about half-way through the cycle. The average peak inventory is $25 million for positive cycles and $21 million for negative cycles.[20]

Figure 2 illustrates the aggregate dealer inventory, as a percentage of bond's issue size, and the mean/median cumulative abnormal return, normalized to zero on the peak inventory date, during positive (Panel A) and negative (Panel B) cycles for a sample of investment grade, large issues with age of one year or more. We calculate bond return as the change in volume weighted average price between two trading days, and abnormal return as the bond return less Bank of America Merrill Lynch's Investment Grade Corporate Bond Index return.

For positive cycles, Figure 2, Panel A shows the build-up in dealer inventory during load phase and the reduction in dealer inventory during unload phase. The abnormal returns are negative during load phase, which reflects the decline in bond prices in response to sustained customer selling, and positive during unload phase, which reflects the reversal in bond prices that compensates liquidity suppliers. For negative cycles, Panel B shows that the abnormal returns are positive during load phase, which reflects the increase

[19] For an individual dealer, Schultz (2017) finds that inventory position is mean-reverting, and the half-life of inventory is about four weeks for actively traded investment grade bonds and five weeks for high yield bonds. Our inventory cycles reflect the cumulative position of all individual dealer positions in a bond.

[20] Summary Statistics Table at the end of Appendix 1 presents cycle length and peak inventories aggregated by year. Inventory cycles remain large and significant throughout the sample but are relatively shorter and shallower in recent years. These results support that dealer capital has declined over the sample period (see, for example, Bessembinder et al. (2018), Schultz (2017), Bao, O'Hara, and Zhou (2018), Dick-Nielson and Rossi (2019), Choi and Huh (2019) and Friewald and Nagler (2019)).

in bond prices in response to sustained customer buying. In contrast to positive cycles, the reversal in bond prices during unload phase is small, indicating that negative cycles are largely associated with permanent price effects. Consistent with these asymmetric patterns, Cai et al. (2019) find evidence of buy-sell price impact asymmetry of institutional herding.[21] Edmans, Levit, and Reilly (2017) argue that bond purchases convey more information than bond sales because buyers have more flexibility in selecting a bond while sellers select the bonds that they already own, due to short sale constraints.

Table 2 shows a similar pattern for the full sample of inventory cycles.[22] During the build-up (load) phase, the returns are negative for positive cycles and positive for negative cycles. The returns have the opposite sign for the unload phase of inventory cycles. For the full inventory cycle, the cumulative returns are marginally negative for positive cycles and significantly positive for negative cycles. We exploit the asymmetry of bond returns in positive and negative cycles to investigate whether funds that we identify as liquidity supplying are sophisticated or naïve participants.

Prior research has shown that the downgrade of a bond from investment grade to speculative grade (Ellul et al. (2011)) and poor past bond performance (Cai et al. (2019)) are associated with a surge in sustained customer selling. Figure 2, Panels C and D show that both events are associated with large increases in the aggregate dealer inventory and the fraction of bonds in positive cycles. These trends persist for over a month and gradually decline.

IV. Classifying bond funds based on trading style

To measure trading style, we overlay the change in a fund's bond holdings on the inventory cycle in a bond. We classify the holdings change as liquidity supplying if the fund trades in the same direction as dealers, i.e., holding change is positive (negative) when inventory cycle is positive (negative), and the opposite is classified as liquidity demanding. Stated differently, a holding change is classified as liquidity supplying (demanding) when a fund purchases (sells) a bond during a period when dealers face sustained customer

[21] Bao, Pan, and Wang (2011) report that price reversals are stronger after price reductions than price increases.
[22] We calculate returns as percentage changes in bond's clean price from the beginning of the cycle to the peak for the buildup phase and from the peak to the end of the cycle for the unloading phase.

selling (buying). Appendix 2 illustrates the advantages of classifying trading style using inventory cycles versus other measures, such as TRACE trade imbalances or bond returns.

The market structure of corporate bonds offers several ways for institutions to gauge the liquidity conditions in a bond. Dealers "signal" their interest in unwinding positions by broadcasting indicative bids and offers on a list of bonds on their inventory (called "Runs") to institutional clients. In the early part of our sample period, dealers used to broadcast runs once a day but in recent years, runs are updated several times intra-day using automated pricing models. In addition, institutions subscribe to feeds from data aggregators (e.g., Bloomberg) for recently completed transactions. They also receive quotations from electronic bond platforms, solicit quotations from dealers via electronic request-for-quote (RFQ) platforms (such as MarketAxess), obtain "color" on market conditions by speaking with dealers, and participate in all-to-all trading platforms that allow institutions to respond to RFQs.[23]

The mechanism of liquidity provision by bond institutions that we have in mind is as follows. During a positive cycle, dealers signal their large positive positions by lowering indicative quotes. A fund with a liquidity supplying trading style uses the opportunity to buy the bond for its portfolio, or avoids selling the bond, both of which alleviate dealer stress. Da et al. (2011) note that active fund managers have a natural advantage in making a market in stocks that they cover, as their superior information about the stock allows them to identify profitable trading opportunities. One notable result from Da et al. (2011) is that income-oriented equity funds earn better returns from liquidity supply, and thus are more likely to engage in such strategy. Given the similarities between income-oriented equity funds and corporate bond funds, liquidity supply could be a value-added strategy in the case of bond funds.

After classifying each bond holding change as liquidity supplying, demanding or unclassified, we calculate the *LS_score* for the fund-reporting period, as follows:

[23] The market share of all-to-all platforms even towards the end of our sample period is negligible. Bank-affiliated dealers are the primary liquidity providers in RFQ platforms.

$$LS\_score = \frac{Liquidity\ supplied\ (\$) - Liquidity\ demanded\ (\$)}{Liquidity\ supplied\ (\$) + Liquidity\ demanded\ (\$) + Unclassified\ (\$)} \quad (1)$$

We require a minimum overlap of 50% between the fund's reporting window and the inventory cycle, and classify bond holdings changes that do not coincide, or significantly overlap, with inventory cycle in a bond as "unclassified."[24] We eliminate holdings changes that overlap with primary bond issuance, as our focus is on liquidity provision in the secondary market. Both the liquidity supplying and liquidity demanding activities of a fund affect the *LS_score*.[25] From equation (1), a fund that neither absorbs nor strains dealer inventories has an *LS_score* of zero, while a fund with a positive *LS_score* exhibits a trading style that on average helps alleviate large dealer positions.

Table 3, Panel A reports descriptive statistics for *LS_score* over our sample period. For the 49,020 fund-period observations[26], the pooled mean and median values of *LS_score* are about -0.05. Aggregating at fund level, the mean *LS_score* estimated over 1,206 bond funds is -0.039 and the median score is -0.049. Thus, the typical mutual fund exhibits a trading style that strains large dealer positions. The average *LS_score* varies over time, and notably, decreases during the financial crisis, from -0.027 in 2007 to -0.082 in 2008, before reversing to -0.045 in 2010. Overall, trading style shows large variation, with the 25th percentile of -0.165 and the 75th percentile of 0.063.

Table 3, Panel B, presents results on trading style persistence. We partition funds into quintiles based on trading style over a rolling 12-month period.[27] During the ranking period, the average *LS_score* ranges from -0.180 for liquidity demanding funds (Q1) to 0.088 for liquidity supplying funds (Q5). In a future, non-overlapping 12-month period [*t+1, t+12*] (and even further in period [*t+13, t+24*], which we do not report for brevity), the average *LS_score* increases monotonically from Q1 to Q5 funds.

[24] That is, for a holdings change to be classified in the case of one-month (three-month) reporting period, the reporting window and cycle must overlap for at least 15 (45) days. The overlap requirement of 50% ensures that each position change is classified in only one way. As reported in Table 3, about half of the holding changes are not classified.
[25] The use of liquidity supply and demand in dollars places greater weight on larger holdings changes, which we believe is an appropriate reflection of a fund's willingness to provide liquidity. A measure based on number of liquidity supplying and demanding holdings changes (i.e., equally weighting each holding change) yields similar results.
[26] Only fund-period observations with complete fund characteristics (to be used as controls in subsequent regressions) and at least four TRACE position changes (used in the calculation of *LS_score*) are included.
[27] We do not observe a difference in the distribution of fund's reporting frequency across trading style quintiles.

We present a transition matrix that reports the overlap between fund rankings in months [*t-11, t*] and fund rankings in non-overlapping future months [*t+1, t+12*]. If past rankings are simply noise, then funds have an equal (20%) probability of being assigned to a *LS_score* quintile in future months. We find that Q5 Funds have a 31.27% probability of being placed in future Q5 quintile and only 13.44% probability of being placed in future Q1 quintile. Similarly, Q1 Funds have a 29.79% probability of being placed in future Q1 quintile, and only 11.60% probability of being placed in future Q5 quintile. The Pearson's chi-square test rejects the null that past trading style has no information about future trading style.

In terms of economic significance, Q5 funds are 20% of all funds in each time period (by definition) and represent about 14% of total TNA, on average, over the sample period. In addition, as a conservative estimate of the economic importance, we identify a subset of funds with average *LS_score* > 0 over the life of the fund in our sample. Based on this classification, liquidity supplying funds, on average, represent about 13% by number and 10% by TNA of all funds over the sample period.

In a test of liquidity supply being sophisticated versus naïve (*Hypothesis H1*), we calculate the future *LS_score* separately using only trades that overlap with positive or negative cycles.[28] Persistence in trading style is associated with positive cycles alone. Based on positive cycles, future *LS_score* rises monotonically from Q1 to Q5 funds and the Q5-Q1 difference is positive and statistically significant. Notably, future *LS_score* for Q4 and Q5 funds is greater than zero, indicating that the trading style of these funds help alleviate dealer stress in future positive cycles. In contrast, based on negative cycles, future *LS_score* does not show a monotonic pattern and is negative for all *LS_score* quintiles. Together, the univariate statistics provide preliminary evidence that our classification of trading style captures the decision of a sophisticated fund to supply liquidity around price pressure events.[29]

[28] Specifically, in equation (1), *LS_score* from positive (negative) cycles is calculated using only the dollar amounts of liquidity supplied and liquidity demanded for bonds in positive (negative) cycles.

[29] Choi and Huh (2019) study the riskless principal trades where dealers act as brokers and match two customers. The study notes that matching trades are consistent with liquidity provision by customers instead of dealers. Unfortunately, TRACE data do not provide the identity of a customer, and as a result, Choi and Huh (2019) are unable to examine systematic behavior by any individual customer. Although pre-arranged trades are more common in recent years, Bessembinder et al. (2018) report that they still represent only a small slice (less than 10%) of overall volume.

We note that data limitations introduce noise in classifying trading style. TRACE corporate bonds account for only 37% of total position changes in a fund-reporting period (Table 1), and about 50% of corporate bond position changes are not classified (Table 3). Our assumption is that trading style that we estimate based on fund's transactions in corporate bonds is, though noisy, informative about its overall trading style. Further, even though the majority (72%) of fund-reporting periods are at a monthly frequency, position changes that occur within the reporting window are not captured in our classification. For equity markets, Puckett and Yan (2011) show that the interim trades of institutions contain useful information about trading skill. Both the low frequency of fund reporting and the transactions in non-TRACE eligible bonds introduce noise in the measurement of trading style. These data limitations reduce the power of our tests to detect the relation between trading style and fund outcomes.

V. Trading style and returns

By monitoring liquidity conditions and incorporating them into trading decisions, fund managers have an opportunity to outperform their peers by both earning a compensation for liquidity supply and/or lowering the price impact of their trades around transitory price pressure events. *Hypothesis H2* predicts that funds that engage in liquidity supply earn additional risk-adjusted returns. To test the hypothesis, we estimate performance by the fund's alpha relative to a four-factor benchmark model, following Chen and Qin (2017), as follows:

$$R_{i,t} - R_{ft} = \alpha_i + \beta_{i,STK} STK_t + \beta_{i,BOND} BOND_t + \beta_{i,DEF} DEF_t + \beta_{i,OPTION} OPTION_t + \varepsilon_{i,t} \quad (2)$$

where $R_{i,t}$ is the return of fund i in month t, R_{ft} is the risk-free rate as measured by one-month Treasury bill rate, *STK* is the stock market factor calculated as the excess return on the CRSP value-weighted stock index, *BOND* is the bond market factor calculated as the excess return on the U.S. aggregate bond index, *DEF* is a measure of default risk premium calculated as the return spread between the high-yield bond index and the intermediate government bond index, and *OPTION* is the option factor calculated as the return spread between the GNMA mortgage-backed security index, which contains prepayment options, and the intermediate government bond index. We estimate the betas using monthly observations over a rolling 18-month period [*t-17, t*]. The beta estimates are then used to calculate the expected return in month *t+1*. The difference between the actual fund return and the expected return yields the estimated alpha for month *t+1*.

We examine whether a fund's liquidity supply is associated with alphas using the following model:

$$\propto_{i,[t+1,t+12]} = \beta_{LS} LS\_scoreQ_{i,[t+1,t+12]} + \beta_{LS-M} LS\_scoreQ_{i,[t+1,t+12]} * MktLiq_{[t+1,t+12]} + \sum_{k=1}^{n} \beta_{X,k} X_{k,i,t} + \sum T_t + \sum FC_i + \varepsilon_{i,[t+1,t+12]} \quad (3)$$

where $\propto_{i,[t,+1\ t+12]}$ is fund i's average alpha over a 12-month period, $LS\_scoreQ_{i,[t+1,t+12]}$, the explanatory variable of interest, is the fund's quintile rank of average LS_score over the same 12-month period, and $LS\_scoreQ_{i,[t+1,t+12]} * MktLiq_{[t+1,t+12]}$ represents the interaction between trading style and measures of market stress. The two measures of market stress are *Crisis*, the 12-month average of a crisis dummy which equals one if a month falls in the period from July 2007 to April 2009, and the 12-month average VIX index.[30] The observable fund attributes $X_{k,i,t}$ are as of the end of month *t* and include measures of investor base stability (including institutional share fraction and rear load), variables capturing asset allocation and liquidity (number of fund holdings, cash holdings and squared cash holdings, government bond holdings, average bond issue size, and average credit rating), and other important fund characteristics that may be related to alpha (broker-dealer affiliated dummy, fund total net assets, and fund age). Following the literature, we also include the fund's lagged flows and returns in the each of the most recent 12 months. Finally, we include month fixed effects denoted by $\sum T_t$ to absorb unobserved time-varying market-wide forces, and fund-category fixed effects denoted by $\sum FC_i$ to absorb time-invariant factors specific to fund categories, and double-cluster the standard errors by fund family and month.

The results in Table 4 show that liquidity supply is positively associated with fund alphas. Model

[30] Due to the rolling 12-month periods used in the estimation, *Crisis* is constructed to measure the overlap between the 12-month period and the crisis period. For example, *Crisis* would equal 3/12 if three of the 12 months fall within July 2007 and April 2009. As robustness, we also find similar results using a *Crisis* dummy variable that equals one if the beginning or ending month of the 12-month period falls within the crisis period.

(1) regresses funds' alphas on contemporaneous *LS_score* quintile rank, along with the control variables and fixed effects. The coefficient on *LS_score* quintile rank is positive and significant at the 1% level. The coefficient indicates that liquidity supplying funds (Q5) outperform liquidity demanding funds (Q1) by two basis points per month (0.482 x 4). To provide some perspective, the average monthly gross alpha of our sample funds is 3.40 basis points.

In model (2), we separately examine the contribution of liquidity supply to fund alpha in positive and negative cycles. We find that fund alpha is associated with trading style calculated over positive cycles, while there is no significant relationship between fund alpha and trading style over negative cycles. Funds in the top quintile of liquidity supply in positive cycles outperform those in the bottom quintile by about three basis points per month (0.796 x 4). Model (3) adds the interactions of trading style in positive and negative cycles with the *Crisis* variable. Model (4) uses the VIX index as a measure of market stress and yields similar results. During periods of market stress, a liquidity supplying trading style calculated over positive cycles is associated with a significantly higher fund alpha while, again, there is no relation between alpha and trading style calculated over negative cycles. In model (5), we include the fund's alpha from the prior period to account for (unobservable) manager skill that might be correlated with trading style and find robust results. Consistent with Chen and Qin (2017), the coefficient on prior-period alpha is positive and significant, which points to performance persistence and evidence of manager skills.

Together, the results from models (1)-(5) provide empirical support for *Hypothesis H2*. We observe a robust relation between trading style and fund performance. The contribution of trading style to alpha is driven by the fund's trading in bonds in positive cycles, which are, on average, associated with transitory price pressure.[31]

For equity funds, Da et al. (2011) find that performance persistence only exists for funds that trade high PIN stocks, which suggests that the ability to process and trade on complex information and not

[31] We explore the contributions to alpha and persistence separately for the load and unload phases of inventory cycles. We find no material difference in the results. While we might expect the return to liquidity provision to be highest when a fund buys or avoids selling close to the peak date of a positive cycle, the granularity of the data does not allow us to accurately conduct such an analysis.

liquidity provision is a source of outperformance. In our setting, the ability to process complex information and trade in a timely manner is likely to be associated with a liquidity demanding trading style. Given that bonds are less information sensitive than stocks and bond liquidity is scarcer, it is useful to know whether the source of performance persistence, or skill, for bond funds is similar to that observed for equity funds.

We split the funds into quintiles by average *LS_score* and study alpha persistence separately for funds in Q1 (i.e., LD funds) and Q5 quintiles (i.e., LS funds). Consistent with the findings for equity funds, model (6) shows that performance persistence exists for LD funds. The alpha difference between funds in highest and lowest past alpha quintiles is about 7 basis points per month. However, unlike equity funds, LS funds exhibit even stronger performance persistence; the alpha difference between highest and lowest past alpha quintiles is larger, at about 12 basis points per month.

VI. What explains trading style?

In this section, we consider *Hypotheses H3* on the role of fund characteristics in hindering or facilitating a strategy to supply liquidity. We split fund characteristics into three groups: measures of funding stability, measures of asset liquidity, and other attributes that relate to organizational aspects of the fund. Table 5, Panel A report estimates from panel predictive regressions of the fund's *LS_score*, in the reporting period beginning at time *t*, on lagged *LS_score* quintile rank and the fund characteristics, as of the end of month *t*:

$$LS\_score_{i,,t+1} = \beta_{LS} LS\_scoreQ_{i,[t-11,t]} + \sum_{k=1}^{n} \beta_{X,k} X_{k,i,t} + \sum T_t + \sum FC_i + \varepsilon_{i,t+1} \qquad (4)$$

All models include time and fund-category fixed effects. Standard errors are double-clustered by fund family and reporting period. Columns (1) to (4) report results for the individual groups and the combined set of fund characteristics, and columns (5) and (6) report the results on trading style persistence.

Stable investor clientele reduces the fund's need to demand liquidity in response to fund flows and has the potential to enhance a fund's ability to engage in liquidity supply (Goldstein et al. (2017)). Funds can lower the volatility of flows by attracting sophisticated investors with longer horizons. Sophisticated investors are more likely to correctly infer the managers' skills from past performance and to be less sensitive to performance volatility (Huang, Wei, and Yan (2012)). Investors with long horizons are less likely to trade in response to short term news and experience liquidity shocks (Cella et al. (2013), Giannetti and Kahraman (2018), and Manconi et al. (2012)). Following the literature, we proxy for investor sophistication and horizon by the proportion of institutional ownership in a fund, the existence of rear loads, and more directly, through revealed preference, past fund flows, the standard deviation of flows, and the past correlation between flow and performance. In model (1), we find that, consistent with stable investor base hypothesis, funds with higher fraction of institutional share classes and lower realized flow-performance sensitivity are more likely to exhibit a liquidity supplying trading style.

In model (2), we examine the impact of portfolio holdings on trading style. Investors in funds with large cash holdings are less sensitive to the liquidation costs imposed by redeeming investors. In light of the trade-off between cash holding's role as a buffer against outflows and the cost of holding a low-yield asset (Chernenko and Sunderam (2018)), we allow for a non-linear effect of cash on trading style. We find that the cash holdings' coefficient is positive, and the squared cash holdings' coefficient is negative. Based on model coefficients, we estimate that cash holdings have a positive effect on the decision to supply liquidity until a threshold of 13.5% of fund's portfolio and a negative effect beyond that threshold (differentiating model (2) with respect to % Cash and equating the result to zero, we find the maximum effect at % Cash = 0.081/(2 x 0.003) = 0.135). Extremely high cash holdings may be indicative of unusual circumstances or other types of strategies being implemented by the fund. Funds that hold smaller issue bonds, which are typically less liquid, exhibit a higher propensity to supply liquidity, which points to participation by LS funds in bonds where dealer interest may be lacking.

In model (3), we examine the impact of fund and fund family attributes on trading style. An affiliated brokerage desk at the fund family could confer informational advantages on market conditions that are difficult for other funds to replicate. Consistent with this idea, the coefficient on broker affiliation dummy is positive.[32] In terms of economic significance, affiliated funds provide more liquidity by about

[32] Many studies on corporate bonds highlight the benefits of a strong dealer network for institutions. O'Hara, Wang, and Zhou (2017) show that less active institutions receive worse executions than more active institutions. Hendershott, Li, Livdan, and Schurhoff (2019) develop a model and show that larger insurance companies obtain better executions by fostering competition among dealers. The model in Green, Hollifield and Schurhoff (2007) predicts that dealers

0.015 in *LS_score* terms, a quarter of the cross-sectional standard deviation of fund-level average *LS_score.* Younger and smaller funds are more likely to supply liquidity. One explanation is that older and larger funds have complex organizational structures, which lead to a greater separation between portfolio manager and trading desk. With the exception of fund age, which loses statistical significance, the results are similar in model (4), which combines all fund characteristics, explaining 2.4% of the variation in *LS_score*.[33]

In comparison, the explanatory power of a model with fund fixed effects (not reported) is 5.8%, which is more than twice as large as that of model (4). A model with fund fixed effects and combined fund characteristics together raises the explanatory power to 5.9%. These results suggest that while certain fund characteristics facilitate or hinder liquidity supply, the choice of trading style is a strategic decision, consistent with *Hypothesis H1*. Model (5) further shows that trading style is persistent; the loading of current trading style on lagged trading style quintile rank is positive and statistically significant. Notably, the R-squared of model (5) with only lagged trading style rank variable is comparable to that of model (4) with a host of fund characteristics, which points to relative importance of trading style in understanding a fund's trading behavior.

In model (6), after accounting for combined fund characteristics, past trading style is still similarly informative about funds' future behavior. In *LS_score* terms, funds classified in Q5 based on past average *LS_score* (LS funds) have a higher future *LS_score* than those classified in Q1 (LD funds) by about 0.05 (1.158/100 x 4), or 21% of the full sample standard deviation. In model (7), we examine a subsample of funds with at least 50% of portfolio invested in corporate bonds and find similar results. Together, the persistence results confirm that the decision to supply liquidity in response to market imbalances is an important fund-specific attribute.

---

gain monopoly power over customers in fragmented and opaque markets. See also Hollifield, Neklyudov, and Spatt (2017) and Li and Schurhoff (2019) for evidence from structured bond and municipal bond markets.

[33] As an additional analysis of liquidity supplying funds, we study the subset of funds with average *LS_score* > 0 over the life of the fund in our sample. The difference in fund attributes between these funds and others, conditional on fund category and time period, follow a similar pattern to those reported in column (4) of Table 5.

In Table 5 Panel B, we explore alternative explanations for the persistence of trading style, focusing on the impact of fund flows on buying and selling activity. As discussed in Section 2, a fund that receives inflows coincident with many bonds experiencing positive cycles could accidently be classified as liquidity supplying as the fund is more likely to purchase than sell bonds. Over time, those funds whose flows are positively (negatively) correlated with incidents of positive (negative) cycles might also exhibit trading style persistence that is accidental. Therefore, we add contemporaneous flows and their interactions with market condition variables to absorb the potential impact of *accidental* liquidity supply.

In model (1), the contemporaneous fund flow over the reporting period of the dependent variable, [*t, t+1*], shows a strong positive association with trading style. A one standard deviation increase in monthly flow is associated with an increase of 0.009 (0.221 x 0.039) in *LS_score*, which is about 17 percentage points of its pooled mean and four percentage points of its pooled standard deviation. After accounting for contemporaneous flows, past trading style remains economically and statistically significant in explaining future trading style. Interestingly, in the specification with contemporaneous flows, the coefficients on past flows (not tabulated for brevity) become insignificant, suggesting that past flows are important only to the extent that they predict current flows. In model (2), we add the interaction between flows and the quintile rank of past *LS_score* (which is scaled from 0 to 4 for ease of interpretation) and find that the interaction term is positive. In terms of economic significance, the effect of flows for LS funds is about twice as large as that for LD funds. Thus, flows enable liquidity supply by LS funds, perhaps by contributing to cash holdings (e.g., inflows increase cash holdings most directly), but also because flows are persistent, implying that high inflows today predict the continuation of funding in the future.

In model (3), we include the interactions between fund flows and the proportions of bonds that are in positive and negative cycles during the contemporaneous reporting period. The model does not include the proportions by themselves since time fixed-effects subsume these variables. The results are as expected – the interaction coefficient on flows and proportion of bonds in positive cycles is positive, and the interaction coefficient on flows and the proportion of bonds in negative cycles is negative, implying that fund that receive inflows (outflows) when a higher proportion of bonds are experiencing positive (negative) cycle are more likely to be classified as liquidity suppliers. After controlling for measures of accidental liquidity supply, past trading style remains informative about future trading style, with a coefficient that is comparable in magnitude to that in model (5) of Panel A. These results suggest that trading style represents a fund's intentional decision to respond to signals of market imbalances.

To test whether liquidity supply is sophisticated or naïve, we re-estimate model (3) with a decomposition of the dependent variable, *LS_score* [*t, t+1*], into components that are associated with bonds in positive and negative cycles. For positive cycles, the estimates for model (4) show that past trading style is predictive of future trading style while for negative cycles, the coefficient on past trading style in model (5) is not statistically significant. In models (6) and (7), we show that this asymmetry between liquidity supply in positive and negative cycles is robust in the subsample of funds with at least 50% of portfolio invested in corporate bonds. These results support *Hypothesis H1* that the typical LS fund that we identify is a sophisticated participant and makes an intentional decision to supply liquidity in positive cycles.

VII. Trading style and fragility

The results thus far indicate that some bond funds help absorb large dealer positions during periods of sustained customer selling. We next examine whether liquidity supplying funds maintain their trading style during periods of elevated stress. We also identify the potential mechanisms by which liquidity supplying funds help alleviate bond market fragility.

A. Fund level analysis

In Table 6, Panel A, we examine trading style when individual funds experience extreme investor flows. Goldstein et al. (2017) show that poor performance leads to large outflows from the aggregate bond fund sector, which has the potential to trigger fire-sale feedback effects. If LS funds help alleviate and/or avoid straining dealers' risk bearing capacity, even under extreme flows, such fragility concerns would not apply to at least this subset of funds. However, as discussed in motivating *Hypothesis H4*, funds may be forced to alter their trading style under flow pressure, given the need for liquidity preservation.

In the spirit of Coval and Stafford (2007), we sort rolling fund-quarter observations into quintiles by average fund flow during the period, and place observations with extreme inflows and outflows in "Flow Q5" and "Flow Q1", respectively. We focus on a three-month horizon, as it is the shortest common span for monthly and quarterly reporting funds, and a longer horizon, such as one year, may dilute the urgency of forced trading. We interpret extreme flow observations during a quarter as plausibly exogenous events that can strain the typical trading style of the fund.

Table 6, Panel A re-examines whether past trading style is associated with future trading style in the face of large inflows (Flow Q5, models (4)-(6)) and outflows (Flow Q1, models (1)-(3)). We present the results based on the following specification:

$$LS\_score_{i,buy\ or\ sell,[t+1,t+3]} = \beta_{LS} LS\_scoreQ_{i,[t-11,t]} + \sum_{k=1}^{n} \beta_{X,k} X_{k,i,t} + \sum T_t + \sum FC_i + \varepsilon_{i,buy\ or\ sell,[t+1,t+3]} \quad (5)$$

where $LS\_score_{i,buy\ or\ sell,[t+1,t+3]}$ is fund *i*'s average *LS_score* from months *t+1* to *t+3*, calculated using only *sell* transactions in large outflow (Q1) periods and only *buy* transactions in large inflow (Q5) periods. $LS\_scoreQ_{i,[t-11,t]}$ is our trading style quintile rank, based on funds' overall trading in the 12-month period ending in month *t*. The model includes the same fixed effects and control variables as in Table 5.

Results in Panel A suggest that past trading style remains informative about future fund behavior, even when funds engage in forced selling activity in response to large outflows (model (1)) and forced buying activity in response to large inflows (model (4)). In columns (2) and (5), we separately examine LD and LS funds using indicator variables instead of *LS_score* quintile rank. For periods with large outflows (model (2)), the negative Q1 fund coefficient indicates that the selling activity of LD funds relative to benchmark funds (Q2 to Q4) are more likely to strain dealer positions while the positive Q5 fund coefficient indicates that the selling activity of LS funds relative to benchmark funds is less likely to strain dealer positions, and the difference in trading style is significant at 1% level. For periods with large inflows (model (5)), the Q5 fund coefficient is positive indicating that LS funds are more likely than benchmark funds to purchase bonds in positive cycles and less likely to purchase bonds in negative cycles.

In models (3) and (6), we repeat the same analysis for a subsample of bonds that are particularly prone to fragility risk due to their illiquidity. Specifically, we focus on high-yield bonds with issue sizes of less than $1 billion. Results are largely similar with slightly larger coefficient estimates for this subsample than for all bonds.

In Table 6, Panels B and C, we study trading style during periods of market stress when dealers are likely to face capital constraints. In Panel B, market stress is identified as time periods where the proportion of bonds in positive cycles are in the highest quintile, which represent episodes of sustained customer selling activity across many bonds. In Panel C, market stress is identified as periods with high values of the VIX index. The dependent variable represents the trading style associated with selling activity in models (1) to (3) and buying activity in models (4) to (6). This analysis does not condition on individual fund flows.

The results in Panel B identify two mechanisms by which funds with a liquidity supplying trading style help alleviate fragility risk. In terms of selling activity, the positive *LS_score* coefficient in model (1) and the positive Q5 coefficient in model (2) indicate that LS funds are less likely than benchmark funds to sell bonds in positive cycles and more likely to sell bonds in negative cycles. In terms of buying activity, the positive *LS_score* coefficient in model (4) and the positive Q5 coefficient in model (5) indicate that LS funds are more likely than benchmark funds to purchase bonds in positive cycles and less likely to sell bonds in negative cycles. Thus, both the buying and selling activities of LS funds exhibit sensitivity to liquidity conditions which helps alleviate fragility risk by absorbing or not straining large dealer positions. Results are similar for the subsample of illiquid bonds (model (3) and (6)) for which fragility risk is of particular concern. Panel C reports broadly similar results for the selling and slightly weaker results for the buying activity of LS funds during periods with high values of the VIX index.

B. Bond level implications

The results thus far indicate that LS funds' selling activity is less likely to strain dealer positions even when the funds face extreme outflows or during periods with market wide stress. The concerns about fragility raised by the findings of Goldstein et al. (2017) and Cai et al. (2019) suggest that (correlated) fund outflows can translate into selling pressure at the bond level, and the bond market can be fragile due to flow-performance feedback effects. Our fund level findings suggest, however, that LS funds are more sensitive to prevailing market conditions and hence less likely than LD funds to create selling pressure at the bond level when facing large outflows.

To explicitly investigate this conjecture, we follow Edmans et al. (2012) in constructing a flow-induced selling (*FIS*) measure for each bond.[34] The measure captures a naïve approach where funds respond to extreme outflows, defined as flows in the bottom 10% of the sample, by liquidating bonds in proportion of the value of the position in their portfolio. We then aggregate into the bond level the flow-induced selling from all funds in each *LS_score* quintile and normalize the sum by bond issue size. The naïve approach contrasts with strategic selling where a fund avoids selling bonds that are illiquid or face selling pressure. Thus, funds that follow a naïve approach are more likely to strain dealer positions and push bonds into (or keep them in) positive cycles while funds that are sensitive to liquidity conditions are less likely to do so. In Table 7, we examine this prediction using the following bond-month level model:

$$BondInCycle_{i,t+1} = \sum_{k=1}^{3} \beta_{FIS,k} FIS_{k,i,t+1} + \sum_{k=1}^{n} \beta_{X,k} X_{k,i,t} + \sum F_i + \varepsilon_{i,t+1} \quad (6)$$

where $BondInCycle_{i,t+1}$ equals one if bond *i* experiences a positive cycle for at least one week in month *t+1*, and equals zero otherwise. $FIS_{k,i,t+1}$ represents our variables of interest, the contemporaneous flow induced selling of LD (Q1) funds, LS (Q5) funds, and the remaining quintiles grouped together. The control variables $X_{k,i,t}$ include bond characteristics that may be related to trading activity and hence the likelihood of being in a positive cycle (or, in dealer inventory), including the level of mutual fund ownership, issue size, time to maturity, bond age, an indicator variable that equals one for investment grade bonds and zero otherwise, indicator variables for bonds upgraded or downgraded within plus or minus two months, and lagged dependent variable (dummy that equals one if the bond is in a positive cycle in month *t*). $\sum F_i$ refers

[34] A similar measure is examined in Jiang et al. (2019).

to bond-issuer fixed effects that control for other unobserved time-invariant issuer-specific variables.

In model (1), we consider flow-induced selling variables alone without controlling for bond characteristics. The results indicate that *FIS* increases the likelihood of a bond being in a positive cycle but the relationship depends on funds' trading style. The coefficient on *FIS* from LD (Q1) funds is positive and highly significant. Q2-Q4 funds also exert a positive but smaller influence on the probability while the coefficient on *FIS* from LS (Q5) funds is negligible and not statistically significant. Thus, a subset of bond funds that we identify as liquidity supplying manage to meet large outflows in a manner that does not raise significant fragility concerns.

In model (2), we control for bond attributes that may be related to the likelihood of a bond being in a positive cycle. The effects of flow-induced selling remain similar in statistical significance but are smaller in economic significance, suggesting that mutual funds tend to hold bonds with certain characteristics and that these characteristics are directly related to the likelihood of being in dealer inventory.[35] The coefficients in column (2) suggest that, while flow-induced selling by LS funds does not impact positive cycles, an increase of *FIS* from LD funds from zero to 0.4% (mean of non-zero observations, which represent about 12% of the sample) raises the likelihood of a bond being in a positive cycle by about 0.02 (14 percentage points of the unconditional mean of 0.144). The same effect is about 0.01 (0.6% x 1.741) for *FIS* from funds in the middle three quintiles. In models (3) and (4), we replicate the analysis for a sample of high-yield, smaller issue bonds for which fragility risk is of particular concern and find similar results.

In Table 8, we apply the analysis in Table 7 to individual bond returns, a measure of fragility that is not directly affected by our definition of inventory cycles. Specifically, we estimate equation (6) with the bond return in month *t+1* as the dependent variable. For a bond in month *t+1*, the return is calculated as the percentage change in volume weighted average price (VWAP) of the bond from the last day on which

[35] The coefficients of control variables are as expected. Larger issue bonds are more likely to be in positive cycles. On the other hand, we find in Table 5 that the weighted average issue size of bonds in a fund's portfolio is negatively related to *LS_score*. Together, these results are consistent with a supply side explanation that dealers are more likely to supply liquidity in larger and more liquid bonds (Goldstein and Hotchkiss (2020)), as well as with a demand-based explanation that there is greater liquidity demand in larger issues (Choi, Shachar, and Shin (2019)).

there are transactions in month *t* to the last day on which there are transactions in month *t+1*. To ensure that we are picking up the return for month *t+1*, we require that the prices come from the last 10 days of months *t* and *t+1*. The estimation includes control variables similar to Table 7 with the exception that we also include lagged *FIS* measures to allow for return reversals. We also include matched index returns and their interaction with bond maturity to absorb systematic variation in bond prices and allow such variation to vary across the yield curve. We use either Bank of America Merrill Lynch's Investment Grade Corporate Bond Index or Bank of America Merrill Lynch's High Yield Corporate Bond Index, depending on the credit rating of the bond of interest, and measure the index return over the exact same trading days that we use to measure the bond return.

In column (1), *FIS* from funds in all *LS_score* quintiles are associated with negative bond returns. This result is consistent with extreme outflows engendering fire-sale price pressures. In the cross-section, *FIS* of LD funds is associated with a price impact that is more than 2.5 times that associated with *FIS* from LS funds. Column (2) adds lagged *FIS* variables and column (3) additionally includes other controls. The coefficients of (contemporaneous) *FIS* for LD funds remain larger in magnitude than those for LS funds in all three columns; the differences are statistically significant at 10% in column (1) and 5% in columns (2) and (3). The coefficients on lagged *FIS* variables indicate significant reversals two months after the extreme outflows for *FIS* associated with LD funds and funds in the middle group, while the reversals associated with *FIS* from LS funds are statistically insignificant. These results suggest that contemporaneous price impact associated with LD funds contains a significant transitory component, confirming that the impact is partially due to fire-sale price pressure.[36] The lack of reversals associated with *FIS* from LS funds is consistent with these funds avoiding sales in bonds that face temporary price pressure. The permanent price impact may be due to the systematic component of flows being correlated with bond returns (flows are

[36] The evidence on mutual fund fire sales of corporate bonds is mixed. Aslan and Kumar (2018) find that mutual fund fire sales are associated with both price effects and real effects during the financial crisis while Choi et al. (2019), using a different approach for measuring abnormal returns, show that fire-sale price effects disappear except in certain extreme scenarios. Our focus is on understanding the cross section of *FIS*. For mismeasurement to explain our results, it has to be more correlated with *FIS* from LD funds than *FIS* from LS funds and the middle group.

significantly correlated across funds with the average correlation of each fund flow and aggregate flow of about 0.15). If we take only the reversal as a measure of price pressure, an increase of *FIS* from LD funds from zero to 0.4% (mean of non-zero observations, which represent about 12% of the sample) decreases bond return by about 60 basis points (about a quarter of the unconditional standard deviation of 2.42%). The same effect is about 37 basis points (0.6% x 0.612) for *FIS* from funds in the middle three quintiles.

In model (4), we again replicate the analysis for a sample of high-yield, smaller issue bonds, and find similar results. In models (5) and (6), we split the bond-month observations into those associated with inventory cycles in which the (orthogonalized) numbers of liquidity-supplying dealers are in the bottom and top quintiles of the sample.[37] Our objective is to investigate whether funds' trading style has smaller differential impacts on price pressure when many dealers share risk, resulting in less overall stress. With the caveat in mind that the number of participating dealers is endogenous, we find that trading style matters less when many dealers share risk, with the price reversals in column (6) being about two-thirds to three quarters of those in column (5) for *FIS* from LD funds and funds in the middle three quintiles. In both cases, *FIS* from LS funds are not associated with significant price reversals.

VIII. Conclusions

Several commentators have argued that the significant growth in the corporate bond market coupled with the reduction in dealer capital point to a liquidity problem in bond markets. Focusing on bond mutual funds, recent work shows that institutional herding in corporate bonds, particularly on the sell side, is significant and that bond fund investors are more sensitive to bad performance than equity fund investors. Given the relative illiquidity of bond holdings, regulators are concerned that large investor flows out of the bond fund sector and the subsequent fire sale of bonds by these funds pose a substantial threat to financial stability.

[37] The number of participating dealers is likely to be endogenous. Dealers are more likely to use the interdealer market or participate as a broker instead of a dealer when market stress is elevated or the bond is illiquid (see Goldstein and Hotchkiss (2020)). At the same time, these actions help reduce the stress. Since it is difficult to find an instrumental variable for risk sharing, we simply orthogonalize the number of participating dealers in a cycle, our measure of risk sharing, from proxies of cycle stress, including cycle length and peak. The orthogonalized number of dealers is the residual from the regression of number of dealers on logs of cycle length and cycle peak.

The linkage between bond institutions and fragility risk is relatively unstudied. Our paper attempts to fill this gap in the literature. We leverage the OTC bond market structure and the availability of dealer trades in TRACE data to build inventory cycles. Focusing on bond mutual funds, we classify a fund's trading style based on its responses to signals of market imbalances, as reflected in inventory cycles. The typical bond mutual fund exhibits a trading style that strains dealer positions. Trading style varies in the cross-section of funds. Importantly, a subset of bond funds intentionally supply liquidity during episodes of large customer selling (that are associated with significant transitory price effects) and earn a return premium for liquidity services. These funds maintain their relative trading style during periods with large fund outflows and elevated market stress, thus alleviating fragility risk of the bond market. This study offers a nuanced view of the potential threats to bond market stability posed by mutual funds.

Our study highlights buy-side institutions as an important liquidity source and a potential solution to the fragility concerns in the corporate bond market. In light of regulatory changes that caused a decline in dealer capital for market making, the bond markets could benefit from liquidity supply by other types of participants. To tap into, and further encourage, the channel of liquidity supply that we identify, regulators need to remove frictions (for example, the market power exercised by large players in OTC markets) that impede broad participation by institutions. In particular, designers of trading systems need to offer protocols that provide institutional and retail investors with *direct access* to platforms (e.g., MarketAxess' Open Trading platform) and the ability to post limit orders that compete with dealer quotations.

References

Aslan, H. and P. Kumar, 2018, The real effects of forced sales of corporate bonds, *Journal of Monetary Economics*, Vol. 95, 1-17.

Amihud, Y. and H. Mendelson, 1980, Dealership market: Market making with inventory, *Journal of Financial Economics*, Vol. 8, 31–53.

Anand, A., P. Irvine, A. Puckett, and K. Venkataraman, 2012, Performance of institutional trading desks: An analysis of persistence in trading costs. *Review of Financial Studies*, Vol. 2 (25), 557-598.

Anand, A., P. Irvine, A. Puckett, and K. Venkataraman, 2013, Institutional Trading and Stock Resiliency: Evidence from the 2007-09 financial crisis. *Journal of Financial Economics*, Vol. 108, 773-797.

Asquith, P., A. S. Au, P. Covert, P. A. Pathak, 2013, The market for borrowing corporate bonds. *Journal of Financial Economics*, Vol. 107, 155-182.

Bao, J., M. O'Hara, and X. Zhou, 2018, The Volcker Rule and market-making in times of stress, *Journal of Financial Economics*, Vol. 130, 93-113.

Bao, J., J. Pan, and J. Wang, 2011, The illiquidity of corporate bonds, *Journal of Finance*, Vol. 66, 911-946.

Bessembinder, H., W. Maxwell, and K. Venkataraman, 2006, Market transparency, liquidity externalities, and institutional trading costs in corporate bonds. *Journal of Financial Economics,* Vol. 82, 251.

Bessembinder, H., S., Jacobsen, W. Maxwell, and K. Venkataraman, 2018, Capital commitment and illiquidity in corporate bonds, *Journal of Finance*, Vol. 74(4), 1615-1661.

Bessembinder, H., C. Spatt, and K. Venkataraman, 2020, A survey of the microstructure of fixed income markets, *Journal of Financial and Quantitative Analysis*, Vol. 55(1), 1-45.

Cai, F., S. Han, D. Li, and Y. Li, 2019, Institutional herding and its price impact: Evidence from the corporate bond market, *Journal of Financial Economics,* Vol. 131, 139-167.

Cella, C., A. Ellul, and M. Giannetti, 2013, Investors' horizons and the amplification of market shocks, *Review of Financial Studies*, Vol. 26(7), 1607-1648.

Chen, Q., I. Goldstein, and W. Jiang, 2010, Payoff complementarities and financial fragility: Evidence from mutual fund outflows, *Journal of Financial Economics,* Vol. 97, 239-262.

Cheng, S., A. Hameed, A. Subrahmanyam and S. Titman, 2017, Short-term reversals: The effects of past returns and institutional exits, *Journal of Financial and Quantitative Analysis,* Vol. 52, 143-173.

Chen Y., and N. Qin, 2017, The behavior of investor flows in corporate bond mutual funds, *Management Science,* Vol. 63, 1365–1381.

Chernenko, S, and A. V. Sunderam, 2018, Do fire sales create externalities? *Journal of Financial Economics*, forthcoming.

Chevalier, J., and G. Ellison, 1997, Risk taking by mutual funds as a response to incentives, *Journal of Political Economy*, Vol. 105, 1167–1200.

Choi, J., S. Hoseinzade, S. S. Shin, and H. Tehranian, Corporate bond mutual funds and asset fire sales, *Journal of Financial Economics*, forthcoming.

Choi, J., and Y. Huh, 2019, Customer liquidity provision: Implications for corporate bond transaction costs, working paper, https://ssrn.com/abstract=2848344.

Choi, J., O. Shachar, and S. S. Shin, 2019, Dealer liquidity provision and the breakdown of the law of one price: Evidence from the CDS-bond basis, *Management Science*, Vol. 65(9), 3949-4450.

Chordia, T., R. Roll, and A. Subrahmanyam, 2011, Recent trends in trading activity and market quality, *Journal of Financial Economics*, 2011, Vol. 101, 243-263.

Cici, G., and S. Gibson, 2012, The performance of corporate bond mutual funds: Evidence based on security-level holdings, *Journal of Financial and Quantitative Analysis*, Vol. 47, 159-178.

Coval, J., and E. Stafford, Asset fire sales (and purchases) in equity markets, *Journal of Financial Economics*, Vol. 86, 479-512.

Cremers, M., and A. Petajisto, 2009, How active is your fund managers? A new measure that predicts performance, *Review of Financial Studies*, Vol. 22, 3329-3365.

Da, Z., P. Gao, and R. Jagannathan, 2011, Impatient Trading, liquidity provision, and stock selection by mutual funds, *Review of Financial Studies*, Vol. 24, 675-720.

Dick-Nielsen, J. and M. Rossi., 2019, Cost of immediacy of corporate bonds, *Review of Financial Studies*, Vol. 32(1), 1-41.

Duffie, D., 2012, "Market making under the proposed Volcker Rule," Working paper, Stanford University.

Ederington, L. H., W. Guan, and P. K. Yadav, 2015, Dealer spreads in the corporate bond market: Agent versus market-making roles, working paper, https://ssrn.com/abstract=2378000.

Edmans, A., D. Levit, and D. Reilly, 2017, The Effect of diversification on price informativeness and governance, working paper, London Business School.

Edwards, A., L. Harris, and M. Piwowar, 2007, Corporate bond market transaction costs and transparency, *Journal of Finance*, Vol 63(3), 1421-1451.

Ellul, A., C. Jotikasthira, and C. Lundblad, 2011, Regulatory pressure and fire sales in the corporate bond market, *Journal of Financial Economics*, Vol. 101, 596-620.

Financial Stability Oversight Council, 2016, Update on review of asset management products and activities, https://www.treasury.gov/press-center/press-releases/Pages/jl0431.aspx.

Franzoni, F., and A. Plazzi, 2015, Do hedge funds provide liquidity? Evidence from their trades, working paper, University of Lugano and Swiss Finance Institute.

Friewald, N., R. Jankowitsch, and M. Subrahmanyam, 2012, Illiquidity or credit deterioration: A study of liquidity in the US corporate bond market during financial crises, *Journal of Financial Economics*, Vol. 105, 18-36.

Friewald, N. and F. Nagler, 2019, Over-the-Counter market frictions and yield spread changes, *Journal of Finance*, Vol 74(6), 3217-3257.

Giannetti, M., and B. Kahraman, 2018, Open-end organizational structures and limits to arbitrage, *Review of Financial Studies*, Vol. 31, 773-810.

Green, R., B. Hollifield, and N. Schürhoff, 2007, Financial intermediation and the costs of trading in an opaque market, *Review of Financial Studies*, Vol. 20, 275-314.

Grossman, S.J., and M.H. Miller, 1988, Liquidity and market structure, *Journal of Finance*, Vol. 43, 617-633.

Goldstein, I., H. Jiang, and D. Ng, 2017, Investor flows and fragility in corporate bond funds, *Journal of Financial Economics*, Vol. 126(3), 592-613.

Goldstein, M., E. Hotchkiss, and E. Sirri, 2007, Transparency and liquidity: A controlled experiment on corporate bonds, *Review of Financial Studies*, Vol. 20, 235-273.

Goldstein, M., and E. Hotchkiss, 2020, Providing liquidity in an illiquid market: Dealer behavior in U.S. corporate bonds, *Journal of Financial Economics,* Vol. 135(1), 16-40.

Hansch, O., N. Y. Naik, and S. Viswanathan, 1998, Do inventories matter in dealership markets? Evidence from the London Stock Exchange, *Journal of Finance*, Vol. 53, 1623-1656.

Harris, L., 2016, Transaction costs, trade through, and riskless principal trading in corporate bond markets, working paper, University of Southern California.

Huang, J., K. Wei, and H. Yan, 2012, Investor Learning and Mutual Fund Flows, working paper, http://ssrn.com/abstract=972780.

Hendershott, T., D. Livdan, D. Li, and N. Schürhoff, 2019, Relationship trading in OTC markets, Journal of Finance, https://doi.org/10.1111/jofi.12864.

Hendershott, T. and A. Madhavan, 2015, Click or call? Auction versus search in the over-the-counter market, *Journal of Finance*, Vol. 70, 419-447.

Hollifield, B., A. Neklyudov, and C. Spatt, 2017, Bid-Ask spreads, trading networks, and the pricing of securitization, *Review of Financial Studies,* Vol. 30(9), 3048-3085.

International Monetary Fund, Global Financial Stability Report, Lower for Longer, October 2019.

Jiang, H., D. Li, and A. Wang, 2019, Dynamic liquidity management by corporate bond mutual funds, working paper, https://ssrn.com/abstract=2776829.

Jylha, P., K. Rinne, and M. Suominen, 2014, Do hedge funds supply or demand liquidity?, *Review of Finance*, Vol. 18 (4), 1259–1298.

Kraus, A., and H. Stoll, 1972, Price impact of block trading on the New York Stock Exchange, *Journal of Finance*, Vol. 27 (3), 569-88.

Li, D., and N. Schürhoff, 2019, Dealer networks, *Journal of Finance,* Vol. 74(1), 91-144.

Manconi, A, M. Massa, and A. Yasuda, 2012, The role of institutional investors in propagating the crisis of 2007-2008, *Journal of Financial Economics*, Vol. 104, 491-518.

Massa, M., A. Yasuda, and L. Zhang, 2013, Supply uncertainty of the bond investor base and the leverage of the firm, *Journal of Financial Economics*, Vol. 110, 185-214.

Nagel, S., 2012, Evaporating liquidity, *Review of Financial Studies*, Vol. 25, 2005-2039.

O'Hara, M., Y. Wang, and X. Zhou, 2017, Best execution of corporate bonds, *Journal of Financial Economics*, Vol. 130(2), 308-326.

Panayides, M., 2007, Affirmative obligations and market making with inventory, *Journal of Financial Economics*, Vol. 86, 513-542.

Puckett, A., and S. Yan. 2011. The interim trading skills of institutional investors. *Journal of Finance,* Vol. 66, 601–633.

Schultz, P., 2001, Corporate bond trading costs: A peek behind the curtain, *Journal of Finance*, Vol. 56, 677-698.

Schultz, P., 2017, Inventory management by corporate bond dealers, working paper, https://ssrn.com/abstract=2966919

Sirri, E., and P. Tufano, 1998, Costly search and mutual fund flows, *Journal of Finance*, Vol. 53, 1589–1622.

Stoll, H.R., 1978, The supply of dealer services in securities markets, *Journal of Finance*, Vol. 33(4), 1133-1151.

Appendix 1: Inventory cycles

We refer to a cycle during which aggregate dealer inventory is positive (negative) as a positive (negative) inventory cycle. Since inventory cycles reflect the opposite side of customer trading, a positive inventory cycle occurs when there is sustained selling from customers. Our inventory cycle starts and ends at zero inventory. On the start date, the inventory departs from zero in either the positive or the negative direction as a result of new trades. The end date is the date immediately before the inventory crosses zero again.

Simply accumulating inventory over time is prone to several issues: potential reporting errors that may compound infinitely;[38] our inability to separate proprietary positions from market making;[39] and the lack of information on inventory that dealers hold from primary market operations. To minimize the impact of these issues, when the cycle length exceeds three calendar months (63 trading days), we select the rolling window of 63 trading days to allow for the slow build-up and unwinding of inventory in an illiquid market. The 63-day window is broadly consistent with the half-life inventory of about four to five weeks discussed in Duffie (2012) and estimated in Schultz (2017) for an active, individual dealer.

In the first 63 trading days, we calculate the cumulative dealer imbalances (henceforth, dealer inventory) from the beginning of the cycle based on the accumulation of daily customer trade imbalance absorbed by the dealer community. Example 1 provides the simplest case, in which the cycle ends within the first 63 trading days as a result of new trades causing the dealer inventory to cross zero. Examples 2-4 represent the cases where a cycle becomes longer than 63 trading days as the dealer inventory does not revert to zero during this period. For these cases, we drop the daily customer trade imbalance that occurs 63 trading days earlier ("*older*") from the dealer inventory calculation.

Dropping the *older* imbalance can cause a new inventory cycle to begin in the opposite direction.[40] We make an adjustment to starting inventory to counteract the mechanical effect, as explained using a few

[38] An example of reporting "error" is a broker dealer who transfers all risk positions to trading desk of a European subsidiary who is not a FINRA member. These offset affiliated-trades are reported as customer trades on TRACE.
[39] The inventory positions of dealers are typically mean-reverting suggesting that the trade is motivated by liquidity provision. Proprietary positions that are motivated by information are unlikely to show similar patterns. Given that trade motivation is not reported, it is not possible to separate the two types of dealer positions.
[40] As robustness, we construct a simple 63-day cumulative inventory measure (discussed later) and find similar results.

scenarios. In example 2, we consider a cycle that ends on a day without any reported transactions in the bond. That is, the dealer inventory is pushed across zero due to the *older* imbalance being dropped. In this case, we reset the starting inventory of the new cycle to zero. In example 3, a cycle ends on a day with reported transactions in the bond and also *older* imbalance being dropped. The effect of the *older* imbalance is sufficient to push the dealer inventory across zero. In this case, we reset the starting inventory of new cycle to be the reported transactions on the day. In example 4, the combination of the reported transactions in the bond and the *older* imbalance being dropped pushes the dealer inventory across zero. In this case, the starting inventory of the new cycle is the residual of the new trades that is in excess of zero after offsetting the previous day inventory.

Example 1: Short cycles

In the example below, we present several cycles observed during trading days 1 to 83. Besides the first column, which indexes the trading day, the table has seven other columns. Column 2 presents the count of the days that the bond is in an inventory cycle. Column 3 presents the (signed) daily aggregate dealer imbalance, which is the opposite of customer trade imbalance and represents the incremental inventory taken on by dealers on that day. Column 4 presents the cumulative imbalance from start of the cycle. Column 5 presents the 63-trading day rolling sum of daily imbalances. In example 1, the bond starts trading for the first time on trading day 1; therefore, cumulative imbalance and the rolling sum imbalance until day 63 are identical. Column 6 presents our "inventory" measure. Column 7 indicates whether the inventory cycle clears our minimum threshold for inclusion, and if so, whether the cycle is positive or negative. Column 8 reports the cycle length.

We highlight the inventory measure in bold and italics. The column where the inventory is drawn from (either cumulative imbalance in column 4 or rolling sum imbalance in column 5) is highlighted in bold, while the other series is de-emphasized in a lighter font. In example 1, since all the cycles are shorter than 63 trading days, the inventory is always drawn from the cumulative imbalance.

	Cycle ending only by new trade						
Trading Day	Day of Cycle	Daily imbalance	Cumulative Imbalance	63-Day Rolling	***Inventory***	Cycle	Cycle Length
1	1	$12	**$12**	$12	***$12***	Excluded	4
2	2	$12	**$24**	$24	***$24***	Excluded	4
3	3	$0	**$24**	$24	***$24***	Excluded	4
4	4	-$12	**$12**	$12	***$12***	Excluded	4
5	-	-$12	**$0**	$0	***$0***	None	-
…	-	$0	**$0**	$0	***$0***	None	-
19	1	$12	**$12**	$12	***$12***	Positive	46
20	2	$12	**$24**	$24	***$24***	Positive	46
…	3 to 44	$0	**$24**	$24	***$24***	Positive	46
63	45	$0	**$24**	$24	***$24***	Positive	46
64	46	-$12	**$12**	$0	***$12***	Positive	46
65	1	-$20	**-$8**	-$32	***-$8***	Negative	18
66	2	-$12	**-$20**	-$44	***-$20***	Negative	18
67	3	$0	**-$20**	-$32	***-$20***	Negative	18
…	4 to 17	$0	**-$20**	-$20	***-$20***	Negative	18
82	18	$10	**-$10**	-$22	***-$10***	Negative	18
83	-	$10	**$0**	-$24	***$0***	None	-

The first cycle begins on day 1 when the dealers buy $12 million. On day 2, the dealers buy another $12 million, resulting in the cumulative imbalance and the inventory of $24 million. On day 4, a sale of $12 million decreases the cumulative imbalance to $12 million. Another sale of $12 million on day 5 ends the cycle. By our definition, the first cycle ends on day 4 (immediately before crossing zero), and therefore the cycle length is only 4 days. Since we require a minimum cycle length of five days, this cycle does not clear our threshold and is excluded from our analysis.

There is no further trading until day 19 when a new positive cycle begins with the dealers buying bonds worth $12 million. The dealers continue to increase positions with another $12 million on day 20, pushing the cumulative imbalance and the inventory cycle to its peak at $24 million. The inventory remains $24 million until day 64 when two large daily negative imbalances on days 64 and 65 close the positive cycle. We start a new negative cycle on day 65 at -$8 million. The new negative cycle then lasts until day 82 when two large positive imbalances on days 82 and 83 together close the cycle at zero.

Example 2: Long cycle that ends on a day when older imbalance drops out

Cycle ending only by old trade dropping out							
Trading Day	Day of Cycle	Daily imbalance	Cumulative Imbalance	63-Day Rolling	***Inventory***	Cycle	Cycle Length
1	1	$12	**$12**	$12	***$12***	Positive	81
2	2	$12	**$24**	$24	***$24***	Positive	81
3	3	$0	**$24**	$24	***$24***	Positive	81
4	4	$0	**$24**	$24	***$24***	Positive	81
5	5	$0	**$24**	$24	***$24***	Positive	81
…	6 to 18	$0	**$24**	$24	***$24***	Positive	81
19	19	$12	**$36**	$36	***$36***	Positive	81
20	20	$12	**$48**	$48	***$48***	Positive	81
…	21 to 62	$0	**$48**	$48	***$48***	Positive	81
63	63	$0	**$48**	$48	***$48***	Positive	81
64	64	$0	$48	**$36**	***$36***	Positive	81
65	65	$0	$48	**$24**	***$24***	Positive	81
66	66	-$20	$28	**$4**	***$4***	Positive	81
67	67	$0	$28	**$4**	***$4***	Positive	81
…	68 to 81	$0	$28	**$4**	***$4***	Positive	81
82	-	$0	**$0**	-$8	***$0***	None	-
83	-	$0	**$0**	-$20	***$0***	None	-

Example 2 illustrates the case of one long inventory cycle that eventually ends as older imbalances are dropped from inventory. As discussed earlier, when a cycle becomes longer than 63 trading days, we switch the inventory measure from cumulative imbalance to the 63-trading day rolling sum imbalance.

Here, a positive cycle starts on day 1 with the $12 million imbalance. The dealers increase their positions on days 2, 19, and 20, resulting in the peak inventory of $48 million which lasts until day 63. On day 64, we switch the inventory measure to the rolling sum imbalance, and therefore the inventory decreases to $36 million despite no trading. Notice that while the cumulative imbalance remains at $48 million, the rolling sum imbalance decreases to $36 million due to the $12 million imbalance on day 1 being dropped (outside the 63-trading day window). On day 65, the $12 million imbalance on day 2 also gets dropped, pushing the inventory down further to $24 million. On day 66, the dealers sell $20 million, further reducing their inventory to $4 million. On day 82, the cycle ends as the $12 million imbalance on day 19 is dropped, pushing the inventory across the zero line. Since there is no imbalance on day 82, we do not begin a new cycle on the day. Following our convention, the long positive cycle lasts 81 days, and peaks at $48 million.

Once the cycle ends, we reset the cumulative imbalance to zero on day 82, ignoring all trades that are part of the completed inventory cycle, and the inventory equals the cumulative imbalance of zero. In contrast, the rolling sum imbalance on day 82 is -$8 million, which reflects that the $12 million imbalance on day 19 is dropped, more than offsetting the $4 million inventory held on day 81. The rolling sum becomes further negative on day 83 as the $12 million imbalance on day 20 is dropped. Notice here that the rolling sum is never reset and may span across cycles.

Example 3: Long cycle that ends on a day when old trades drop out and new trades occur

Example 3 is a minor variation of Example 2. The key difference is that Example 3 reports an imbalance of $12 million whereas Example 2 reports no imbalance on day 82. In both examples, dropping the older imbalance on day 19 is sufficient to push inventory to cross zero.

	Cycle ending by old trade dropping out; a trade occurring on the same day						
Trading Day	Day of Cycle	Daily imbalance	Cumulative Imbalance	63-Day Rolling	***Inventory***	Cycle	Cycle Length
1	1	$12	**$12**	$12	***$12***	Positive	81
2	2	$12	**$24**	$24	***$24***	Positive	81
3	3	$0	**$24**	$24	***$24***	Positive	81
4	4	$0	**$24**	$24	***$24***	Positive	81
5	5	$0	**$24**	$24	***$24***	Positive	81
…	6 to 18	$0	**$24**	$24	***$24***	Positive	81
19	19	$12	**$36**	$36	***$36***	Positive	81
20	20	$12	**$48**	$48	***$48***	Positive	81
…	21 to 62	$0	**$48**	$48	***$48***	Positive	81
63	63	$0	**$48**	$48	***$48***	Positive	81
64	64	$0	$48	**$36**	***$36***	Positive	81
65	65	$0	$48	**$24**	***$24***	Positive	81
66	66	-$20	$28	**$4**	***$4***	Positive	81
67	67	$0	$28	**$4**	***$4***	Positive	81
…	68 to 81	$0	$28	**$4**	***$4***	Positive	81
82	1	-$12	**-$12**	-$20	***-$12***	Negative	> 2
83	2	$0	**-$12**	-$32	***-$12***	Negative	> 2

Once the cycle ends, we reset the cumulative imbalance on day 82 to zero, and inventory equals the cumulative imbalance of zero. In example 3, we begin a new negative inventory cycle on day 82 to reflect the daily imbalance of -$12 million.

Example 4: Long cycle that ends on a day when old trades drop out and new trades occur

Example 4 is a minor variation of example 3. The key difference is that example 4 reports a daily imbalance of $8 million on day 66, whereas example 3 reports a daily imbalance of $20 million on day 66. Until day 65, example 3 and 4 are the same. On day 66, the sale of $8 million in example 4 is smaller than $20 million in example 3, and therefore decreases the inventory to $16 million. On day 82, due to $12 million older imbalance on day 19 being dropped, the 63-day rolling imbalance decreases from $16 million to $4 million.

	Cycle ending by a combination of new trade and old trade dropping out						
Trading Day	Day of Cycle	Daily imbalance	Cumulative Imbalance	63-Day Rolling	***Inventory***	Cycle	Cycle Length
1	1	$12	**$12**	$12	***$12***	Positive	81
2	2	$12	**$24**	$24	***$24***	Positive	81
3	3	$0	**$24**	$24	***$24***	Positive	81
4	4	$0	**$24**	$24	***$24***	Positive	81
5	5	$0	**$24**	$24	***$24***	Positive	81
…	6 to 18	$0	**$24**	$24	***$24***	Positive	81
19	19	$12	**$36**	$36	***$36***	Positive	81
20	20	$12	**$48**	$48	***$48***	Positive	81
…	21 to 62	$0	**$48**	$48	***$48***	Positive	81
63	63	$0	**$48**	$48	***$48***	Positive	81
64	64	$0	$48	**$36**	***$36***	Positive	81
65	65	$0	$48	**$24**	***$24***	Positive	81
66	66	-$8	$40	**$16**	***$16***	Positive	81
67	67	$0	$40	**$16**	***$16***	Positive	81
…	68 to 81	$0	$40	**$16**	***$16***	Positive	81
82	1	-$12	**-$8**	-$8	***-$8***	Negative	> 2
83	2	$0	**-$8**	-$20	***-$8***	Negative	> 2

In contrast to example 3, the dropping of older imbalance from day 19 does not end the inventory cycle. Here, the cycle ends only because the -$12 million daily imbalance on day 82 further pushes the inventory to -$8 million (+$4 million - $12 million). The positive cycle ends on day 81. A new negative cycle starts on day 82 with the residual, -$8 million, as the starting inventory.

Robustness: Rolling Sum as Alternative Inventory Measure

As a robustness analysis, we construct inventory cycle using the rolling sum of imbalances (column 5) over 63 trading days. The rolling sum is easier to understand, and the cycles are similar in length and peak inventory to the main approach. The main results are similar to those reported in the paper. A disadvantage of the rolling sum imbalance is that dropping an older imbalance could start a new cycle in the opposite direction. As an illustration, in example 2, the 63-day rolling sum imbalance would start a new negative cycle on day 82. Such a mechanical cycle is dropped in the main approach.

Summary Statistics

The following table provides the summary statistics for cycle length (days) and inventory peak ($ million) by year to supplement Table 2:

	Positive Inventory Cycle (N = 117,825)			Negative Inventory Cycle (N = 90,046)			Diff.
	Mean	Std. Dev.	Median	Mean	Std. Dev.	Median	Mean
Cycle length by year (Full cycle - Days)							
2003	79.723	58.882	77.000	86.924	60.050	91.000	-7.201***
2004	74.279	57.642	66.000	79.908	58.797	80.000	-5.628***
2005	78.817	59.219	76.000	80.738	57.935	84.000	-1.921*
2006	77.748	58.679	71.000	80.016	58.047	79.000	-2.269**
2007	80.681	59.744	80.000	81.982	59.017	83.000	-1.301
2008	76.822	59.922	71.500	87.129	62.924	90.000	-10.307***
2009	66.466	55.816	50.000	75.060	57.898	68.000	-8.594***
2010	72.913	54.764	68.000	73.671	55.646	69.000	-0.759
2011	73.250	56.537	69.000	75.467	56.865	70.000	-2.217**
2012	65.811	49.901	63.000	65.907	51.303	57.000	-0.095
2013	63.577	47.990	62.000	60.986	45.522	56.000	2.591***
2014	64.647	48.746	63.000	63.002	47.183	58.000	1.645**
2015	61.508	47.137	57.000	61.774	46.773	58.000	-0.267
2016	59.483	46.696	54.000	55.456	43.116	47.000	4.026***
2017	64.947	49.724	63.000	60.264	46.156	53.000	4.683***

Cont'd next page

	Positive Inventory Cycle (N = 117,825)			Negative Inventory Cycle (N = 90,046)			Diff.
	Mean	Std. Dev.	Median	Mean	Std. Dev.	Median	Mean
							Cont'd from previous page
Peak inventory by year ($ Million)							
2003	27.682	22.007	19.946	25.092	18.518	18.871	2.590***
2004	27.028	21.838	19.523	25.077	18.837	18.680	1.950***
2005	26.460	21.375	19.059	24.741	18.769	18.244	1.720***
2006	26.957	21.922	19.379	24.162	17.989	18.292	2.795***
2007	27.930	22.103	20.617	24.241	18.275	18.288	3.689***
2008	23.950	19.723	17.838	21.638	17.204	16.449	2.312***
2009	25.032	20.411	18.188	21.522	16.819	16.316	3.510***
2010	27.591	22.041	20.280	22.915	17.298	17.555	4.676***
2011	26.600	21.544	19.430	22.110	16.595	17.237	4.489***
2012	24.903	20.461	18.335	18.804	14.341	15.087	6.099***
2013	23.549	19.219	17.608	19.257	14.378	15.397	4.292***
2014	23.031	18.787	17.411	18.912	14.363	15.139	4.119***
2015	22.201	18.426	16.581	17.508	12.952	14.514	4.693***
2016	23.064	18.981	17.258	18.190	13.560	14.826	4.874***
2017	25.454	20.221	18.792	19.898	14.254	15.910	5.556***

Appendix 2: Classifying Trading Style Based on Inventory Cycles

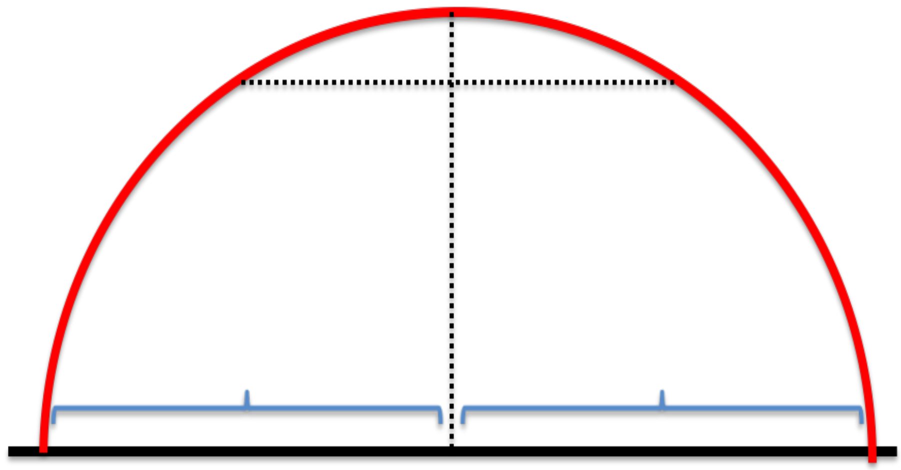
The corporate bond market is relatively illiquid, and the majority of corporate bonds do not trade on a regular basis. Past research shows higher incidences of institutional herding that are associated with large price impact and slow price reversals in bonds than that observed in equities (Cai et al. (2019)). For these reasons, slow-moving dealer inventory positions is an important feature of bond markets that is less relevant for the relatively liquid equity markets.

Our approach is specifically designed to measure trading style of institutions in bond markets and differs from approaches that are suitable for equity markets. Some studies in equity literature classify trading style based on the correlation between fund transactions and daily stock return (Anand et al. (2013), and Cheng, Hameed, Subrahmanyam, and Titman (2017)), or daily TAQ trade imbalance (Da et al. (2011)). Other studies classify trading style based on exposure of the funds' return to a contrarian long-short factor portfolio (Nagel (2012)). We classify the trading style of bond funds based on dealer inventory cycles (IC), which leverage the OTC bond market structure and the detailed TRACE data fields with dealer identifiers. Specifically, the key advantage is that, unlike equity datasets, the inventory positions of key intermediaries (dealers) in bond markets can be measured directly using TRACE dataset. Dealer inventory cycles present a direct and accurate picture of liquidity conditions in bond markets. Inventory cycles also facilitate a cleaner classification of a bond fund's trading style since both bond returns and trade imbalances exhibit opposing patterns during the load and unload phases of the cycle.

The figure below illustrates the advantages of classifying trading style using dealer IC instead of TRACE customer trade imbalance (ΔI), or bond returns (R_b). Suppose a fund purchases a bond when dealers in aggregate face selling pressure from customers. The purchase implies that the fund's change in bond holdings (ΔH) between two reporting periods is positive. The sign of dealer inventory cycle (IC) is positive, which reflects aggregate positive dealer inventory that is driven by customer selling. In contrast, the sign of trade imbalance and bond returns, both have opposite signs in the load and unload phases of inventory cycle. Specifically, in the load phase, the customer selling activity which initiates the cycle leads to a negative TRACE imbalance and negative bond returns while in the unload phase, the customer buying activity that ends the cycle leads to positive TRACE imbalance and positive bond returns. The extent of price reversal depends on the temporary and permanent price impact of customer-driven trade imbalances in the bond.

Under our approach, when correlation between change in fund holdings and sign of inventory cycle (ΔH, IC) is positive, we classify the transaction as liquidity supplying, since the fund is purchasing the bond from dealers when other customers in aggregate are selling the bond. Our classification is not sensitive to whether the fund's reporting period overlaps with load or unload phase of cycle. In contrast, the correlation between change in fund holdings and TRACE trade imbalance (ΔH, ΔI), or the correlation between change in fund holdings and bond returns (ΔH, R_b) have the opposite sign in load or unload phase of positive cycle.

Our approach is subject to some limitations that introduce noise in classifying trading style. First, only a subset of a fund's transactions between reporting periods are used to classify a fund's trading style. Specifically, transactions in bonds that are not TRACE-eligible are not classified. Second, similar to other approaches, we lack detailed data on the exact timing of transactions, which limits our ability to test theoretical predictions; for example, whether return to liquidity provision is highest when a fund buys or avoids selling close to the peak date of a cycle, as predicted by inventory models. As noted in footnote 31, the differences in our main findings are not striking when we classify transactions based on the extent of overlap between reporting period and the load and unload phases of a cycle; nonetheless, the low-frequency data limits our ability to reach a definitive conclusion.



	Load Phase		Unload Phase
TRACE trade imbalance (ΔI)	-		+
Bond return (r)	-		+
Inventory cycle (IC)		+	
Change in fund holdings (ΔH)		+	
Correlation (ΔH,ΔI)	-		+
Correlation (ΔH,r)	-		+
Correlation (ΔH,IC)	+		+
Trading style	**Liquidity supply**		**Liquidity supply**

Figure. Classifying the change in fund's holding as liquidity supply during a positive inventory cycle.

Table 1
Summary Statistics for Taxable Bond Funds

This table presents summary statistics for fund characteristics (Panel A) and turnover (Panel B) of taxable bond funds. The data are from Morningstar, and the sample period is from July 2003 to June 2017. The sample includes only open-ended funds in Morningstar classifications with average corporate bond allocation of 30% or greater: Corporate Bond, High-Yield Bond, Multisector Bond, Nontraditional Bond, Bank Loan, Preferred Stock, Short-Term Bond, Intermediate-Term Bond, and Long-Term Bond. The observation frequencies are fund-month for flow and return, and fund-month (family-month) or coarser, depending on each fund's reporting frequencies, for other variables at the fund (family) level. Number of positions is the number of unique bond CUSIPs held by each fund on each report date. Flows and returns are measured as a percentage of prior-month total net assets (TNA) while the allocations to cash and equivalents, corporate bonds, government bonds, and others (including municipal bonds, securitized bonds, and derivatives) are measured as a percentage of current-month TNA. TRACE positions are positions in TRACE sample bonds. Average credit rating is the value-weighted averages of bonds' credit rating (1 = AAA, 2 = AA+, etc.), calculated using the worst credit rating among S&P, Moody's and Fitch, and only corporate bonds for which the credit ratings are available through Mergent FISD. Index fund dummy equals one if Morningstar classifies the fund as index fund, and zero otherwise. Institutional share fraction is the fraction of TNA that is owned by institutional share classes. Rear load is the value-weighted average across all share classes of the maximum charge, in percentage points, for redeeming the mutual fund shares. Family definition is as reported by Morningstar, and the total net assets and number of funds reported here only include taxable bond funds. Total (TRACE) position change is the par value of changes in all (TRACE) bond positions. Turnover is the annualized ratio of total position change and prior-reporting date TNA. The statistics are reported by length of time in months between two reporting dates.

Panel A: Fund Characteristics

	N	Mean	Std. Dev.	Pct. 25	Median	Pct. 75
Total net assets (TNA, $ Million)	72,632	1,873	7,916	106	377	1,234
Number of positions	72,632	463	868	145	276	493
Asset allocation (%)						
Cash	72,632	9.12	14.97	3.02	6.39	12.40
Corporate bonds	72,632	50.10	28.92	26.03	43.44	79.86
Government bonds	72,632	14.34	15.97	0.01	9.80	23.25
Others	72,632	26.44	23.49	6.89	24.75	41.68
TRACE positions/TNA (%)	72,632	40.44	30.28	16.50	31.15	68.51
Average credit rating	72,632	9.58	4.83	6.00	8.00	15.00
Institutional share fraction (%)	72,632	38.83	41.16	0.00	20.88	85.10
Rear load (%)	72,632	0.33	0.75	0.00	0.00	0.18
Index fund dummy	72,632	0.03	0.18	0.00	0.00	0.00
Monthly return (%)	98,707	0.40	1.19	-0.17	0.39	1.05
Monthly flow (%)	98,707	0.52	3.89	-1.18	0.05	1.64
Family total net assets ($ Million)	20,585	8,990	36,227	148	763	5,638
Number of funds in family	20,585	6	7	1	3	7

Table 1 *-continued*

Panel B: Fund Trading

	N	Mean	Std. Dev.	Pct. 25	Median	Pct. 75
Total pos. change ($ Million)	72,632	412	1,069	13	55	226
TRACE pos. change ($ Million)	72,632	61	124	3	13	53
TRACE/Total pos. change (%)	72,632	37.03	33.05	8.31	25.24	66.82
Turnover by reporting period (annualized)						
1 month	52,422	2.25	2.29	0.80	1.51	2.76
2 months	1,506	2.61	3.56	0.94	1.55	2.68
3 months	18,704	1.90	1.70	0.87	1.41	2.28
All reporting periods	72,632	2.16	2.19	0.83	1.48	2.61

Table 2
Summary Statistics for Dealer Inventory Cycles

This table presents summary statistics for characteristics of dealer inventory cycles. Each inventory cycle begins when the cumulative inventory of all dealers' changes from zero and ends when it comes back to zero. A positive (negative) inventory cycle is a cycle during which the cumulative inventory is positive (negative), i.e. dealers buying more than selling (buying less than selling) in aggregate. For each bond on each trading day, cumulative inventory is calculated using all customer trades from the beginning of the cycle if the beginning of the cycle is less than three months ago, or over the past three months if the beginning of the cycle is more than three months ago. Bond trading data, including trade size, trade price, and whether the trade is between two dealers or between dealer and customer, are from TRACE, and the sample period is from July 2003 to June 2017. Cycle length is the number of calendar days in an inventory cycle. Loading (unloading) period is the period over which the cumulative inventory moves away from (back to) zero. Peak inventory is the largest cumulative inventory, most positive or most negative in par value terms, during the cycle. Bond return between two trading days is calculated using volume-weighted average price (VWAP) of all trades. Only cycles with peak inventory of $10 million or greater and with cycle length of 5 days or longer are included. Tests of difference in mean between the positive and negative inventory cycles are conducted using heteroscedasticity-robust standard errors. *, **, and *** refer to statistical significance at 10%, 5%, and 1% levels.

	Positive Inventory Cycle (N = 117,825)			Negative Inventory Cycle (N = 90,046)			Diff.
	Mean	Std. Dev.	Median	Mean	Std. Dev.	Median	Mean
Cycle length (Days)							
Loading	32.6	34.1	20	34.4	32.2	24	-1.678***
Unloading	35.2	29.7	27	35.5	30.8	27	-0.242
Full	68.8	53.3	64	70.6	53.9	65	-1.865*
Peak inventory ($ Million)	25.0	20.4	18.3	21.2	16.3	16.3	3.864***
Bond return (%)							
Loading	-0.164	2.562	-0.026	0.524	2.817	0.096	-0.688***
Unloading	0.152	2.584	0.043	-0.045	2.716	-0.022	0.197***
Full	-0.025	3.789	-0.001	0.473	4.164	0.077	-0.499***

Table 3
Summary Statistics for Funds' Liquidity Supply and Its Persistence

This table presents summary statistics for mutual funds' liquidity supply measure (Panel A) and its persistence (Panel B). Observations are fund-month or coarser, depending on each fund's reporting frequency. Liquidity supply score (*LS_score*) is calculated as:

$$LS\_score = \frac{Liquidity\ supplied\ (\$) - Liquidity\ demanded\ (\$)}{Liquidity\ supplied\ (\$) + Liquidity\ demanded\ (\$) + Unclassified\ (\$)}$$

For each fund-bond-period, the fund is considered "supplying" ("demanding") liquidity if the change in the fund's position in that particular bond is on the same (opposite) side as the dealer inventory cycle, and the overlap between the fund's reporting period and the dealer inventory cycle is at least half of the fund's reporting period. Changes in the fund's position in a bond that coincide with the bond's initial public offering are excluded. Changes in the fund's positions that do not have the required overlap with inventory cycles are considered "unclassified." Changes in par value are then aggregated across all corporate bonds, grouped into liquidity supplied, liquidity demanded, and unclassified. The three aggregate changes, for each fund-period, are used in the above calculation. *LS_score* is winsorized at 2.5% in both tails. In Panel A, the statistics for *LS_score* are calculated for the entire sample (pooled, counting each observation as one unit) and for the cross section of funds' time-series averages, both over the full sample period and in each calendar year. The last column reports the mean fraction, in par value terms, of unclassified position changes. In Panel B, funds are sorted into five quintiles by their *LS_score*, averaged over the period from months *t*-11 to *t*. The first column reports the sorting variable. The next three columns report the average *LS_score* for funds in each quintile over the next twelve months (*t*+1 to *t*+12), with the numerator of the above calculation reflecting, respectively, all position changes, only changes that coincide with positive inventory cycles, and only changes that coincide with negative inventory cycles. The last five columns report the likelihood of a fund in an *LS_score* quintile during the sorting period falling into each *LS_score* quintile in the subsequent period *t*+1 to *t*+12. The reported chi-square statistics are for the test of null hypothesis that the probability for being in each *LS_score* quintile in the future is independent of the fund's *LS_score* quintile today. Only fund-period observations with complete fund characteristics (to be used as controls in subsequent regressions) and at least four TRACE position changes (used in the calculation of *LS_score*) are included. Tests of difference in mean between the top and bottom and between the top and middle *LS_score* quintiles are conducted using heteroscedasticity-robust standard errors. $*$, $**$, and $***$ refer to statistical significance at 10%, 5%, and 1% levels.

Table 3 *-continued*

Panel A: Summary Statistics of LS_score

	N	Mean	Std. Dev.	Pct. 25	Median	Pct. 75	Mean Unclass'd Fraction
Pooled	49,020	-0.050	0.228	-0.165	-0.048	0.063	0.498
Fund average	822	-0.045	0.057	-0.080	-0.049	-0.016	0.500
Fund average by year							
2003	382	-0.043	0.179	-0.120	-0.041	0.035	0.419
2004	414	-0.060	0.156	-0.130	-0.062	-0.007	0.444
2005	429	-0.039	0.148	-0.101	-0.037	0.019	0.459
2006	457	-0.029	0.141	-0.110	-0.037	0.030	0.480
2007	472	-0.040	0.142	-0.103	-0.039	0.017	0.476
2008	504	-0.078	0.142	-0.149	-0.071	0.000	0.446
2009	516	-0.063	0.117	-0.116	-0.063	-0.015	0.521
2010	518	-0.042	0.105	-0.100	-0.055	0.002	0.507
2011	538	-0.050	0.119	-0.102	-0.057	-0.003	0.499
2012	563	-0.038	0.107	-0.098	-0.046	0.009	0.551
2013	583	-0.060	0.101	-0.105	-0.061	-0.016	0.550
2014	597	-0.056	0.103	-0.111	-0.057	-0.003	0.540
2015	616	-0.047	0.103	-0.095	-0.043	0.004	0.575
2016	606	-0.058	0.106	-0.106	-0.057	-0.005	0.512
2017	615	0.002	0.178	-0.061	0.000	0.077	0.566

Table 3 *-continued*

Panel B: Future Means and Transition Matrix of LS_score

Quintile of Avg. *LS_score* [*t*-11, *t*]	Avg. *LS_score* [*t*-11, *t*]	Avg. *LS_score* [*t*+1, *t*+12]			Percentage in Quintile of Avg. *LS_score* [*t*+1, *t*+12]				
		Total	Positive Cycles	Negative Cycles	1 (Low)	2	3	4	5 (High)
1 (Low)	-0.180	-0.072	-0.024	-0.047	29.79	23.34	18.87	16.40	11.60
2	-0.097	-0.060	-0.019	-0.041	18.41	27.21	24.74	19.26	10.38
3	-0.051	-0.049	-0.007	-0.042	13.82	24.48	26.33	22.80	12.57
4	-0.004	-0.039	0.010	-0.049	13.38	19.96	23.72	25.54	17.39
5 (High)	0.088	-0.021	0.044	-0.065	13.44	15.56	16.74	22.99	31.27
5 - 1	0.269***	0.051***	0.068***	-0.018***	H0: Rows and Columns are Independent				
5 - 3	0.140***	0.029***	0.051***	-0.022***	$\chi^2 > 3{,}100$***				

Table 4

Funds' Liquidity Supply and Performance

This table reports OLS estimates for panel regressions of alpha (in basis points) on quintile ranks of contemporaneous *LS_score. LS_score* from positive inventory cycles, and *LS_score* from negative inventory cycles. Both the alpha and the liquidity supply measures are averaged over twelve-month rolling periods. Quintile rank of 5 (1) indicates the highest (lowest) average *LS_score*. Observations are fund-months, and only those with at least four identifiable CUSIPS traded per reporting period during months *t*-11 to *t* are included. Alpha is calculated by subtracting benchmark return from actual fund return:

$$R_{i,t+1} - R_{ft+1} = \propto_{t+1} + \left[\beta_{i,STK} STK_{t+1} + \beta_{i,BOND} BOND_{t+1} + \beta_{i,DEF} DEF_{t+1} + \beta_{i,OPTION} OPTION_{t+1} \right]$$

where *STK* is the excess return on the CRSP value-weighted stock index, *BOND* is the excess return on the U.S. aggregate bond index, *DEF* is the return spread between the high-yield bond index and the intermediate government bond index, and *OPTION* is the return spread between the GNMA mortgage-backed security index and the intermediate government bond index. All bond indices are from Bank of America Merrill Lynch downloaded from DataStream. The parameters, β_{STK}, β_{BOND}, β_{DEF}, and β_{OPTION}, are estimated on a rolling basis. For alpha in month *t*+1, the estimation period is from months *t*-17 to *t*. *LS_score*, *LS_score* from positive inventory cycles, and *LS_score* from negative inventory cycles are as defined in Table 3. Crisis is the contemporaneous twelve-month average of crisis dummy, which equals one in the period from July 2007 to April 2009, and zero otherwise. VIX is the CBOE implied volatility index, also averaged in the same manner. Quintile rank of average alpha is calculated by sorting fund into quintiles by their alpha, averaged over the past twelve months (*t*-11 to *t*). Quintile rank of 5 (1) indicates the highest (lowest) average past alpha. All other control variables are as of the end of last month *t*, i.e., the beginning of the current twelve-month rolling period. Institutional share fraction is the fraction of TNA that is owned by institutional share classes. Rear load is the value-weighted average across all share classes of the maximum charge, in percentage points, for redeeming the mutual fund shares. ln(Number of holdings) is natural log of the number of CUSIPs. % Cash is fund's percentage allocations to cash and equivalents. Average credit rating are the value-weighted averages of bonds' credit rating (1 = AAA, 2 = AA+, etc.). ln(Average bond issue size) is natural logs of value-weighted average issue size (in dollars) of corporate bonds in fund's portfolio. Broker affiliated is a dummy variable that equals one if the fund's family is affiliated with a broker-dealer bank, and zero otherwise. ln(TNA) and ln(Age) are natural logs of fund's TNA and age (in years). All continuous variables are winsorized at 2.5% in both tails. All models also include individual monthly flows and returns for months *t*-11 to *t*, and fund classification and month fixed effects. Columns (1)-(5) are for the full sample. Columns (6)-(7) are for the sample of funds whose average *LS_score* quintiles equal 1 and 5, respectively. Standard errors, two-way clustered by fund family and month, are in parentheses. $*$, $**$, and $***$ refer to statistical significance at 10%, 5%, and 1% levels.

Table 4 *-continued*

Sample:	Full Sample					Avg. *LS_score* Q1	Avg. *LS_score* Q5
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Quintile of avg. *LS_score*	0.482***				0.427***		
	(0.150)				(0.142)		
Quintile of avg. *LS_score* from pos. cycles		0.796***	0.872***	0.861***			
		(0.213)	(0.217)	(0.209)			
Quintile of avg. *LS_score* from neg. cycles		0.072	0.064	0.082			
		(0.223)	(0.222)	(0.238)			
Quintile of avg. *LS_score* from pos. cycles x Crisis			2.169**				
			(1.015)				
Quintile of avg. *LS_score* from neg. cycles x Crisis			-0.637				
			(1.030)				
Quintile of avg. *LS_score* from pos. cycles x VIX				0.094*			
				(0.053)			
Quintile of avg. *LS_score* from neg. cycles x VIX				-0.006			
				(0.054)			
Quintile of avg. past alpha [*t*-11, *t*]					2.001***	1.669**	2.895***
					(0.558)	(0.734)	(0.613)
Control variables							
Institutional share fraction	1.669*	1.675*	1.746**	1.750**	1.229	0.706	1.423
	(0.904)	(0.897)	(0.879)	(0.866)	(0.931)	(1.515)	(1.342)
Rear load	0.870	0.849	0.837	0.861	0.795	1.136	0.900
	(0.566)	(0.569)	(0.558)	(0.564)	(0.516)	(0.848)	(0.611)
ln(Number of holdings)	0.930	0.991	1.012	1.023	1.008	1.985**	0.754
	(0.645)	(0.648)	(0.640)	(0.646)	(0.625)	(0.935)	(0.815)
% Cash	0.065	0.062	0.056	0.066	0.048	-0.035	0.120
	(0.104)	(0.102)	(0.098)	(0.099)	(0.095)	(0.093)	(0.101)
(% Cash)2	-0.001	-0.001	-0.001	-0.001	-0.000	0.001	-0.000
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)

Cont'd next page

Table 4 *-continued*

Sample:	Full Sample					Avg. *LS_score* Q1	Avg. *LS_score* Q5
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
% Government	-0.065	-0.069	-0.075	-0.073	-0.060	-0.121	-0.089*
	(0.065)	(0.064)	(0.062)	(0.061)	(0.061)	(0.085)	(0.046)
ln(Average bond issue size)	-0.470	-0.421	-0.423	-0.435	-0.445	0.464	-0.621
	(0.444)	(0.449)	(0.444)	(0.446)	(0.415)	(0.810)	(0.733)
Average credit rating	-0.396*	-0.387*	-0.389*	-0.388*	-0.400*	-0.835**	-0.902***
	(0.218)	(0.221)	(0.222)	(0.223)	(0.211)	(0.408)	(0.272)
Broker affiliated	0.288	0.330	0.476	0.374	0.055	1.643	0.215
	(0.975)	(0.963)	(0.907)	(0.937)	(0.898)	(2.066)	(1.489)
ln(TNA)	0.140	0.142	0.151	0.145	0.074	0.144	0.324
	(0.428)	(0.427)	(0.418)	(0.418)	(0.430)	(0.509)	(0.747)
ln(Age)	0.352	0.385	0.386	0.381	0.289	-0.222	1.228
	(0.689)	(0.682)	(0.683)	(0.691)	(0.637)	(0.967)	(0.968)
Other controls	12 individual lags (*t*-11, *t*) of fund flows and returns						
Time fixed effects	YES	YES	YES	YES	YES	YES	YES
Category fixed effects	YES	YES	YES	YES	YES	YES	YES
Observations	72,794	72,794	72,794	72,794	72,794	13,404	12,812
R-squared	0.175	0.176	0.180	0.181	0.189	0.195	0.224

Table 5

Determinants of Funds' Liquidity Supply

Panel A reports OLS estimates for panel predictive regressions of a fund's liquidity supply measure, *LS_score*, in the next reporting period (beginning at time *t*) on the quintile rank of the fund's average *LS_score* in the past twelve months and the fund's funding, asset, and other general characteristics. Observations are fund-reporting period (unbalanced), and only those with at least four identifiable CUSIPS traded per reporting period during months *t*-11 to *t* are included. At the end of period *t*, funds are sorted into five quintiles by their *LS_score*, averaged over *t*-11 to *t*. Quintile rank of 5 (1) indicates the highest (lowest) average *LS_score*. *LS_score* is calculated as described in Table 3. Funds' characteristics are as of the end of period *t*. Average and standard deviation of flow, as well as the correlation between flow and lagged alpha, are calculated over *t*-11 to *t*. All other variables are defined in Table 4. The first six columns report estimates for the full sample. The last column reports results for the subsample of funds whose average allocations to corporate bonds are at least 50%. Panel B reports OLS estimates for the same model as in column (6) of Panel A but also includes contemporaneous (to the dependent variable) explanatory variables (flow, its interaction with the quintile rank of the fund's average *LS_score*, and its interactions with the fractions of bonds in positive and negative cycles). A bond is considered in a positive (negative) cycle if it is in a positive (negative) inventory cycle for at least one calendar week in the month. Fraction of bonds in positive (negative) cycles is calculated as the number of bonds in positive (negative) cycles divided by the number of bonds that are outstanding at the beginning of the month. In columns (4) and (6) ((5) and (7)), the dependent variable, *LS_score*, is replaced by *LS_score* from positive (negative) inventory cycles. The first five columns report estimates for the full sample. The last two columns report results for the subsample of funds whose average allocations to corporate bonds are at least 50%. All continuous variables are winsorized at 2.5% in both tails. All models include fund classification and month fixed effects. Coefficients are scaled by 100. Standard errors, two-way clustered by fund family and period, are in parentheses. ∗, ∗∗, and ∗∗∗ refer to statistical significance at 10%, 5%, and 1% levels.

Panel A: Predictive Regressions

	Full Sample						Average Corp. ≥ 50%
(Coef. x 100)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Quintile of avg. LS-score [*t*-11, *t*]					1.325***	1.158***	0.927***
					(0.229)	(0.202)	(0.210)
<u>Funding</u>							
Institutional share fraction	1.162**			0.776*		0.705*	0.534
	(0.476)			(0.445)		(0.396)	(0.572)
Rear load	0.246			0.153		0.127	0.464*
	(0.208)			(0.196)		(0.157)	(0.280)
Avg. flow [*t*-11, *t*]	0.426***			0.398***		0.321***	0.275***
	(0.117)			(0.097)		(0.082)	(0.104)
Std. dev. flow [*t*-11, *t*]	0.043			-0.110		-0.146*	-0.074
	(0.111)			(0.095)		(0.088)	(0.165)

Cont'd next page

Table 5 *-continued*

(Coef. x 100)	Full Sample (1)	(2)	(3)	(4)	(5)	(6)	Average Corp. ≥ 50% (7)
Corr. flow and lagged return [*t*-11, *t*]	-0.953***			-0.977***		-0.910***	-0.614
	(0.333)			(0.352)		(0.332)	(0.551)
Assets							
ln(Number of holdings)		-0.228		0.155		0.082	0.362
		(0.293)		(0.290)		(0.239)	(0.391)
% Cash		0.081**		0.070**		0.062**	0.081**
		(0.035)		(0.032)		(0.027)	(0.039)
$(\% \text{ Cash})^2$		-0.003***		-0.003***		-0.002***	-0.003***
		(0.001)		(0.001)		(0.001)	(0.001)
% Government		-0.005		-0.005		-0.004	-0.009
		(0.023)		(0.024)		(0.022)	(0.025)
ln(Average bond issue size)		-0.399*		-0.313*		-0.252	-0.273
		(0.203)		(0.180)		(0.154)	(0.182)
Average credit rating		-0.027		-0.035		-0.042	-0.063
		(0.075)		(0.070)		(0.063)	(0.070)
Fund and family characteristics							
Broker affiliated			1.450**	1.387**		1.166**	1.492*
			(0.630)	(0.572)		(0.493)	(0.833)
ln(TNA)			-0.352**	-0.453***		-0.392***	-0.514***
			(0.149)	(0.125)		(0.103)	(0.169)
ln(Age)			-0.908***	-0.384		-0.318	-0.258
			(0.333)	(0.283)		(0.232)	(0.302)
Time fixed effects	YES	YES	YES	YES	YES	YES	YES
Category fixed effects	YES	YES	YES	YES	YES	YES	YES
Observations	49,020	49,020	49,020	49,020	49,020	49,020	26,442
R-squared (total)	0.022	0.020	0.022	0.024	0.025	0.028	0.034

Table 5 *-continued*

Panel B: Modified Predictive Regressions with Contemporaneous Variables Added and LS-score Decomposed into Those Coming from Positive and Negative Cycles

	Full Sample					Average Corp. ≥ 50%	
Dependent Variable:	*LS_score*	*LS_score*	*LS_score*	*LS_score* Pos. Cycles	*LS_score* Neg. Cycles	*LS_score* Pos. Cycles	*LS_score* Neg. Cycles
(Coef. x 100)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Quintile of avg. *LS_score* [*t*-11, *t*]	1.162***	1.134***	1.110***	1.359***	-0.229	1.119***	-0.225
	(0.200)	(0.200)	(0.198)	(0.245)	(0.157)	(0.244)	(0.144)
Flow x Quintile of avg. *LS_score* [*t*-11, *t*]		0.032**	0.033*	0.037*	-0.009	0.042*	-0.005
		(0.016)	(0.017)	(0.020)	(0.024)	(0.023)	(0.027)
<u>Contemporaneous variables</u>							
Flow	0.221***	0.152**	0.159***	0.867***	-0.597	0.925***	-0.637
	(0.052)	(0.061)	(0.052)	(0.361)	(0.446)	(0.366)	(0.472)
Flow x Fraction of bonds in pos. cycles			6.505***	5.154***	0.751	3.114***	0.352
			(0.919)	(0.982)	(1.190)	(0.965)	(1.341)
Flow x Fraction of bonds in neg. cycles			-10.948***	-9.603***	-1.944	-8.008**	-4.737
			(1.474)	(3.646)	(2.981)	(3.836)	(3.255)
Controls	As in Column (6) of Panel A						
Time fixed effects	YES	YES	YES	YES	YES	YES	YES
Category fixed effects	YES	YES	YES	YES	YES	YES	YES
Observations	49,020	49,020	49,020	49,020	49,020	26,442	26,442
R-squared (total)	0.030	0.030	0.033	0.257	0.237	0.246	0.237

Table 6
Liquidity Supply in Periods of Stress

This table reports OLS estimates for panel predictive regressions of funds' average *LS_score* on funds' quintile rank of past average *LS_score* for the subsamples of fund-period observations associated with large individual fund flows (Panel A) and during periods of market stress (Panels B and C). In all panels, average *LS_score*, the dependent variable, is calculated over the period from months t+1 to t+3 for only sell (columns (1)-(3)) and buy (columns (4)-(6)) transactions, both in the numerator and denominator of the formula. In columns (3) and (6), *LS_score* is calculated using only illiquid bonds, defined as high-yield bonds with issue size less than $1 billion. Past average *LS_score* is calculated over the period from months t-11 to t, using both buy and sell transactions, as defined in Table 5. In Panel A, columns (1)-(3) ((4)-(6)) are for the subsamples of funds in the top (bottom) flow quintile. In Panels B and C, the periods of market stress are defined as the three-month periods (for which we calculate the dependent variable) in which the average fractions of bonds in positive cycles and the average VIX, respectively, are in the top quintile. Panels B and C do not condition on individual fund flows. All control variables (omitted for brevity) are defined in Tables 4-5 and are calculated at the end of last month t, i.e., the beginning of the current three-month period. All continuous variables are winsorized at 2.5% in both tails. All models include fund classification and month fixed effects. Standard errors, two-way clustered by fund family and period, are in parentheses. In columns (2), (3), (5), and (6), reported F-statistics are for the test of null hypothesis that the coefficients of *LS_score* Q1 and *LS_score* Q5 dummies are equal. ∗, ∗∗, and ∗∗∗ refer to statistical significance at 10%, 5%, and 1% levels.

Panel A: Liquidity Supply among Funds Facing Large Outflows and Inflows.

Sample:	Flow Q1 (Large Outflows)			Flow Q5 (Large Inflows)		
Dependent Variable:	*LS_score* (t+1 to t+3, ***sells***)			*LS_score* (t+1 to t+3, ***buys***)		
	All	All	Illiquid	All	All	Illiquid
(Coef. x 100)	(1)	(2)	(3)	(4)	(5)	(6)
Quintile of avg.	1.127***			0.589**		
LS_score [t-11, t]	(0.370)			(0.279)		
[A] Avg. *LS_score*		-2.558**	-3.407**		-0.898	-0.912
[t-11, t] Q1		(1.261)	(1.629)		(0.748)	(1.657)
[B] Avg. *LS_score*		2.805**	3.430*		1.600*	2.110*
[t-11, t] Q5		(1.198)	(2.020)		(0.902)	(1.211)
Controls	Same as in Column (5) of Table 5 Panel A			Same as in Column (5) of Table 5 Panel A		
Category fixed effects	YES	YES	YES	YES	YES	YES
Time fixed effects	YES	YES	YES	YES	YES	YES
H_0: [A] = [B]	NA	F(1, 5585) = 9.68***	F(1, 4561) = 9.00***	NA	F(1, 6023) = 3.43*	F(1, 5126) = 3.12*
Observations	9,178	9,178	6,901	9,519	9,519	7,489
R-squared (total)	0.081	0.081	0.082	0.257	0.257	0.144

Table 6 *-continued*

Panel B: Liquidity Supply during Periods in Which Large Fractions of Bonds Are in Positive Cycles.

Dependent Variable:	*LS_score* (t+1 to t+3, ***sells***)			*LS_score* (t+1 to t+3, ***buys***)		
	All	All	Illiquid	All	All	Illiquid
(Coef. x 100)	(1)	(2)	(3)	(4)	(5)	(6)
Quintile of avg. *LS_score* [t-11, t]	1.730***			0.255*		
	(0.367)			(0.146)		
[A] Avg. *LS_score* [t-11, t] Q1		-4.019***	-3.757***		-0.366	-0.099
		(1.365)	(1.361)		(0.363)	(1.298)
[B] Avg. *LS_score* [t-11, t] Q5		3.028***	2.646*		0.724**	1.011
		(0.953)	(1.582)		(0.310)	(1.772)
Controls	Same as in Column (5) of Table 5 Panel A			Same as in Column (5) of Table 5 Panel A		
Category fixed effects	YES	YES	YES	YES	YES	YES
Time fixed effects	YES	YES	YES	YES	YES	YES
H_0: [A] = [B]	NA	F(1, 3246) = 16.81***	F(1, 2820) = 5.44**	NA	F(1, 3243) = 5.21**	F(1, 2894) = 0.29
Observations	11,365	11,365	8,427	11,353	11,353	8,904
R-squared (total)	0.034	0.032	0.026	0.050	0.050	0.032

Panel C: Liquidity Supply during Periods of High VIX.

Dependent Variable:	*LS_score* (t+1 to t+3, ***sells***)			*LS_score* (t+1 to t+3, ***buys***)		
	All	All	Illiquid	All	All	Illiquid
(Coef. x 100)	(1)	(2)	(3)	(4)	(5)	(6)
Quintile of avg. *LS_score* [t-11, t]	0.715**			0.470		
	(0.357)			(0.332)		
[A] Avg. *LS_score* [t-11, t] Q1		1.091	-1.712		-1.825**	-1.775
		(2.047)	(2.484)		(0.792)	(2.049)
[B] Avg. *LS_score* [t-11, t] Q5		4.152**	4.328**		0.401	2.002
		(1.776)	(2.151)		(1.204)	(2.245)
Controls	Same as in Column (5) of Table 5 Panel A			Same as in Column (5) of Table 5 Panel A		
Category fixed effects	YES	YES	YES	YES	YES	YES
Time fixed effects	YES	YES	YES	YES	YES	YES
H_0: [A] = [B]	NA	F(1, 2447) = 1.27	F(1, 2073) = 3.37*	NA	F(1, 2552) = 2.78*	F(1, 2176) = 1.54
Observations	8,375	8,375	5,861	8,729	8,729	6,197
R-squared (total)	0.026	0.027	0.056	0.124	0.124	0.075

Table 7
Flow-Induced Sales and Positive Inventory Cycles

This table reports OLS estimates for linear models of probability that a bond will be in a positive inventory cycle. Observations are bond-months. In each month *t*+1, all TRACE bonds that exist through the end of the month are included, each as one observation. Dummy for being in a positive cycle, the dependent variable, equals one if a particular bond is in a positive cycle for at least one calendar week in month *t*+1, and zero otherwise. Explanatory variable of interest is flow-induced sales of bond *i* in month *t*+1, *FIS*, by all funds *j* in group *q*, calculated as:

$$FIS_{i,q,t+1} = \frac{\sum_{j \in q} 1_{(flow_{j,t+1} \in bottom 10\%)} \times |flow_{j,t+1}| \times Par_{i,j,t}}{IssueSize_i}$$

where $flow_{j,t+1}$ denotes the flow into fund *j* in month *t*+1, $1_{(flow_{j,t+1} \in bottom 10\%)}$ denotes a dummy that equals one if $flow_{j,t+1}$ is in the bottom 10% of the sample, $Par_{i,j,t}$ denotes the amount of bond *i* held by fund *j* at the end of month *t*, and $IssueSize_i$ is the issue size of bond *i*. Group *q* is defined by quintile of average *LS_score* in the past twelve months (*t*-11 to *t*), where quintiles 2-4 are grouped together for simplicity. Mutual fund ownership is the fraction of bond held by all bond funds at the end of month *t*-1. ln(Bond maturity), ln(Bond issue size), and ln(Bond age) natural logs of maturity (in years), issue size (in dollars) and age (in years) of the bond as of the end of month *t*. Investment grade is a dummy that equals one if the bond is rated BBB- or above by all rating agencies. Upgrade (Downgrade) is a dummy that equals one if the bond is upgraded (downgraded) within two months from the current month. All continuous variables are winsorized at 2.5% in both tails. Columns (1)-(2) are for the full sample, and columns (3)-(4) are for the sample of illiquid bonds, defined as high-yield bonds with issue size less than $1 billion. All models include bond issuer (defined by 6-digit CUSIP) fixed effects. Standard errors, two-way clustered by bond issuer and month, are in parentheses. $*$, $**$, and $***$ refer to statistical significance at 10%, 5%, and 1% levels.

Table 7 *-continued*

	All Bonds		Illiquid Bonds	
	(1)	(2)	(3)	(4)
FIS from fund in avg. *LS_score* [*t*-11, *t*] Q1	22.990***	4.971**	14.899**	4.731*
	(4.603)	(2.447)	(5.550)	(2.638)
FIS from fund in avg. *LS_score* [*t*-11, *t*] Q2 to Q4	5.935**	1.714*	5.209*	1.548
	(2.261)	(0.998)	(2.455)	(0.973)
FIS from fund in avg. *LS_score* [*t*-11, *t*] Q5	-0.140	-3.012	-9.402	-1.979
	(7.464)	(4.546)	(8.898)	(4.026)
Controls				
Positive cycle [*t*]		0.657***		0.697***
		(0.007)		(0.004)
Mutual fund ownership		0.048*		0.034
		(0.023)		(0.022)
ln(Bond maturity)		0.001		0.001
		(0.001)		(0.002)
ln(Bond issue size)		0.040***		0.037***
		(0.002)		(0.002)
ln(Bond age)		-0.025***		-0.023***
		(0.002)		(0.002)
Investment grade		-0.007**		
		(0.003)		
Upgrade		0.019**		0.026***
		(0.007)		(0.004)
Downgrade		0.002		0.041***
		(0.008)		(0.009)
Bond issuer fixed effects	YES	YES	YES	YES
Observations	1,251,009	1,198,446	403,578	376,230
R-squared (total)	0.114	0.521	0.146	0.579

Table 8

Flow-Induced Sales and Bond Returns

This table reports OLS estimates for panel regressions of bond returns on flow-induced sales, *FIS*, and its lags. Observations are bond-months. Columns (1)-(3) are for the full sample, columns (4) is for the sample of illiquid bonds, defined as high-yield bonds with issue size less than $1 billion, and columns (5)-(6) are for the samples of bond-months associated with inventory cycles in which the numbers of liquidity supplying dealers (orthogonalized from the effects of cycle peak and length) are, respectively, in the bottom and top quintiles. For bond *i* in month *t*+1, return is calculated as a percentage change in volume weighted average price (VWAP) of bond *i* from the last day on which there are transactions in month *t* to the last day on which there are transactions in month *t*+1. If the last day on which there are transactions is not within the last 10 days of the month, the two bond-month observations associated with that last day are excluded. *FIS* is calculated as described in Table 7, where the time index *t*+1 denotes the contemporaneous month, and the time indexes, *t* and *t*-1 denote the most recent two lags. For bond *i* in month *t*+1, matched index return is the return in month *t*+1 (over the same dates used to calculate the return on bond *i*) on either Bank of America Merrill Lynch's Investment Grade Corporate Bond Index or Bank of America Merrill Lynch's High Yield Corporate Bond Index, depending on bond *i*'s (worst) credit rating at the end of month *t*. Other control variables are as defined in Table 7. All continuous variables are winsorized at 2.5% in both tails. All models include bond issuer (defined by 6-digit CUSIP) fixed effects. Standard errors, two-way clustered by bond issuer and month, are in parentheses. $*$, $**$, and $***$ refer to statistical significance at 10%, 5%, and 1% levels.

	All Bonds			Illiquid Bonds	No. of Dealers in Q1	No. of Dealers in Q5
	(1)	(2)	(3)	(4)	(5)	(6)
FIS from funds in avg. *LS_score* Q1 [*t*+1]	-1.911**	-2.573**	-2.670**	-3.945***	-3.286**	-1.624*
	(0.783)	(0.923)	(0.921)	(1.153)	(1.371)	(0.892)
FIS from funds in avg. *LS_score* Q2 to Q4 [*t*+1]	-1.178***	-1.248***	-1.266***	-1.998***	-1.644***	-1.088**
	(0.254)	(0.271)	(0.276)	(0.352)	(0.302)	(0.467)
FIS from funds in avg. *LS_score* Q5 [*t*+1]	-0.728**	-0.721**	-0.698**	-1.004**	-0.955***	-0.528
	(0.336)	(0.295)	(0.310)	(0.406)	(0.352)	(0.560)
FIS from funds in avg. *LS_score* Q1 [*t*]		0.492	0.412	0.139	0.726	0.359
		(0.561)	(0.573)	(0.761)	(0.678)	(0.874)
FIS from funds in avg. *LS_score* Q2 to Q4 [*t*]		0.196	0.189	0.184	0.339	0.187
		(0.264)	(0.253)	(0.274)	(0.316)	(0.377)
FIS from funds in avg. *LS_score* Q5 [*t*]		0.214	0.188	0.323	0.132	0.098
		(0.334)	(0.329)	(0.285)	(0.535)	(0.643)

Cont'd next page

Table 8 *-continued*

	All Bonds			Illiquid Bonds	No. of Dealers in Q1	No. of Dealers in Q5
	(1)	(2)	(3)	(4)	(5)	(6)
FIS from funds in avg. *LS_score* Q1 [*t*-1]		1.611**	1.553**	2.476***	2.173**	1.587*
		(0.606)	(0.616)	(0.784)	(0.849)	(0.783)
FIS from funds in avg. *LS_score* Q2 to Q4 [*t*-1]		0.602***	0.612***	1.037***	1.246**	0.826*
		(0.197)	(0.204)	(0.243)	(0.580)	(0.430)
FIS from funds in avg. *LS_score* Q5 [*t*-1]		0.251	0.296	0.467	0.524	0.607
		(0.333)	(0.338)	(0.362)	(0.644)	(0.893)
Controls						
Matched index return [*t*+1]	-0.100	-0.096	-0.095	0.180***	-0.092	-0.035
	(0.081)	(0.083)	(0.084)	(0.035)	(0.082)	(0.051)
Matched index return [*t*+1] x ln(Bond maturity)	0.363***	0.358***	0.357***	0.160***	0.343***	0.370***
	(0.075)	(0.077)	(0.077)	(0.042)	(0.078)	(0.059)
Mutual fund ownership			-0.001	-0.000	-0.001	-0.002
			(0.001)	(0.002)	(0.001)	(0.003)
ln(Bond maturity)			0.001**	0.001	0.000	0.001**
			(0.000)	(0.001)	(0.001)	(0.000)
ln(Bond issue size)			-0.000*	-0.000	-0.000*	-0.000**
			(0.000)	(0.000)	(0.000)	(0.000)
ln(Bond age)			-0.000	-0.000	-0.000	-0.000
			(0.000)	(0.000)	(0.000)	(0.000)
Investment grade			-0.003**	0.000	-0.003**	-0.004***
			(0.001)	(0.000)	(0.001)	(0.001)
Upgrade			0.000	0.001	0.001	0.000
			(0.000)	(0.001)	(0.001)	(0.001)
Downgrade			-0.005***	0.004	-0.004**	-0.007***
			(0.001)	(0.003)	(0.002)	(0.001)
Bond issuer fixed effects	YES	YES	YES	YES	YES	YES
Observations	855,087	828,744	828,744	237,071	92,459	98,204
R-squared	0.307	0.307	0.310	0.257	0.361	0.345

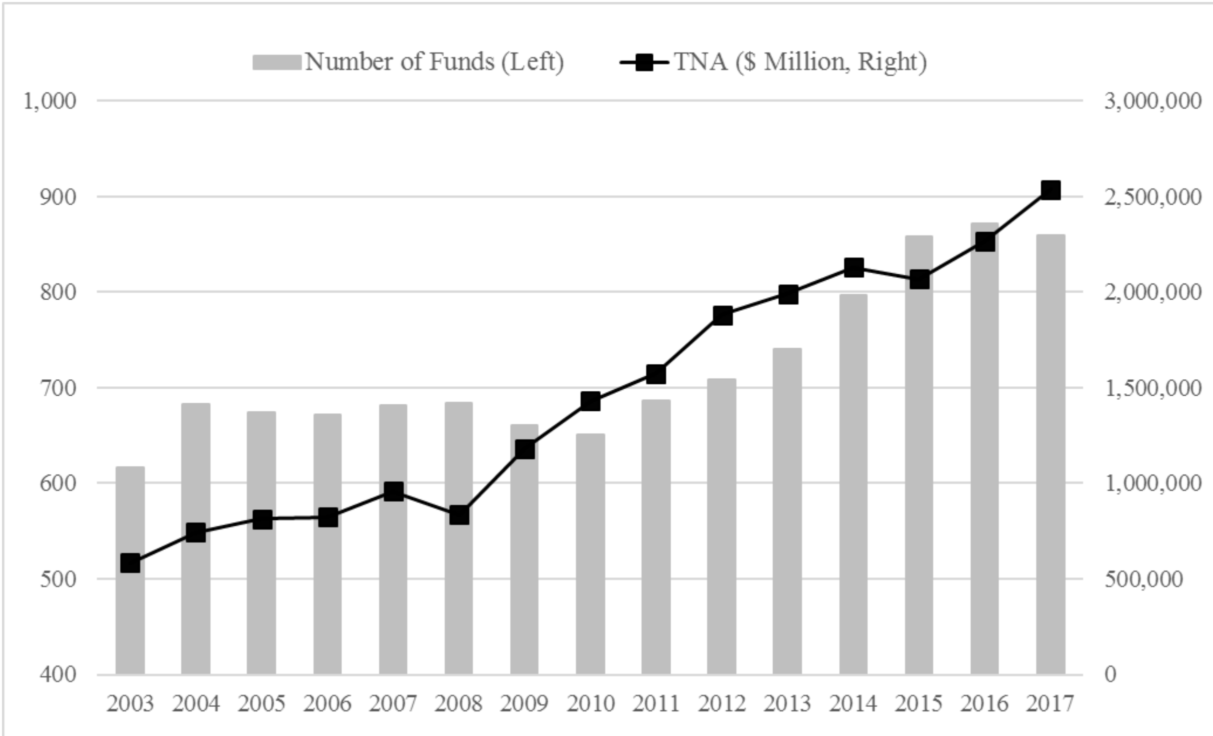


Figure 1. Taxable bond mutual funds over time. This figure presents the total net assets (TNA, in $ Million) of taxable bond funds and the numbers of these funds, as reported by Morningstar, over the sample period from July 2003 to 2017. The sample includes only open-ended funds in the following Morningstar classifications, for which the average allocation to corporate bonds is 30% or greater: Corporate Bond, High-Yield Bond, Multisector Bond, Nontraditional Bond, Bank Loan, Preferred Stock, Short-Term Bond, Intermediate-Term Bond, and Long-Term Bond.

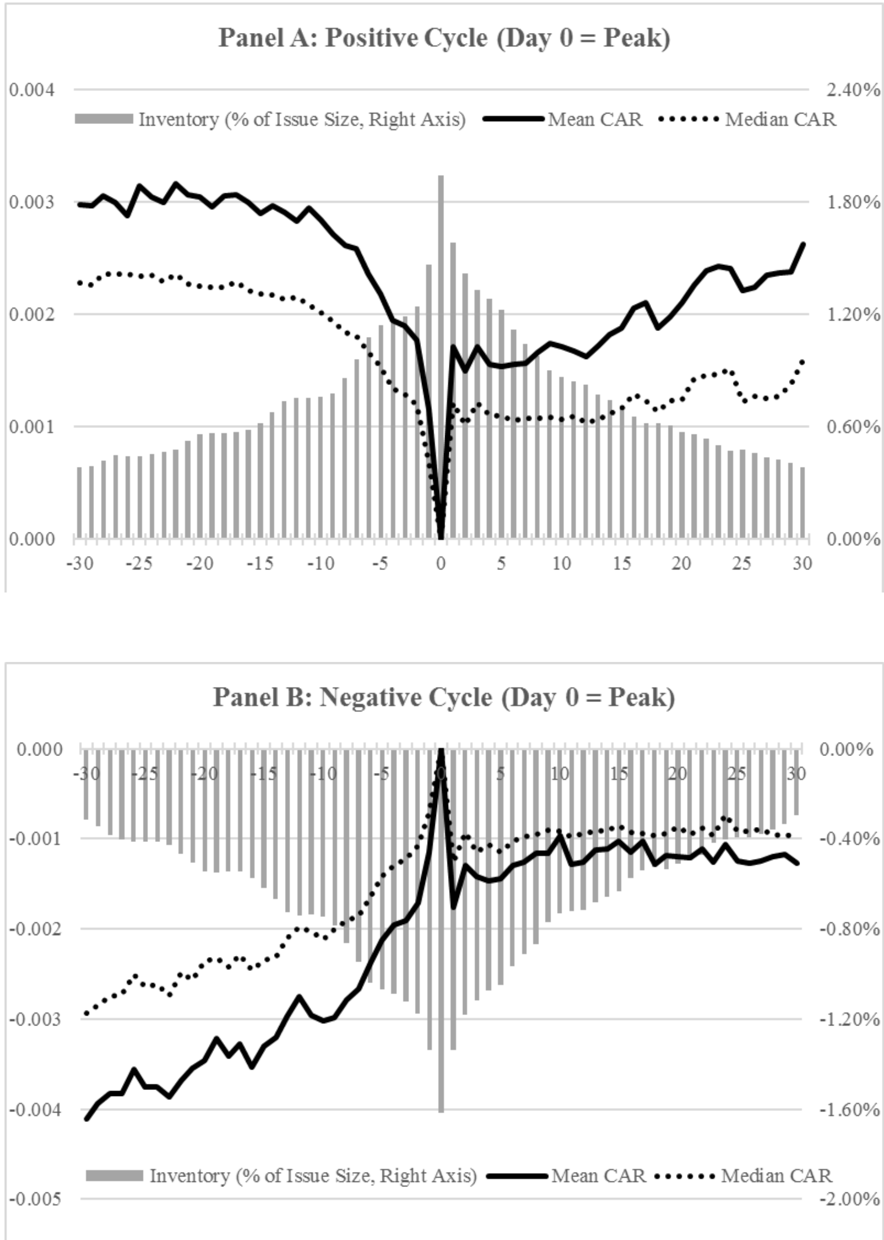
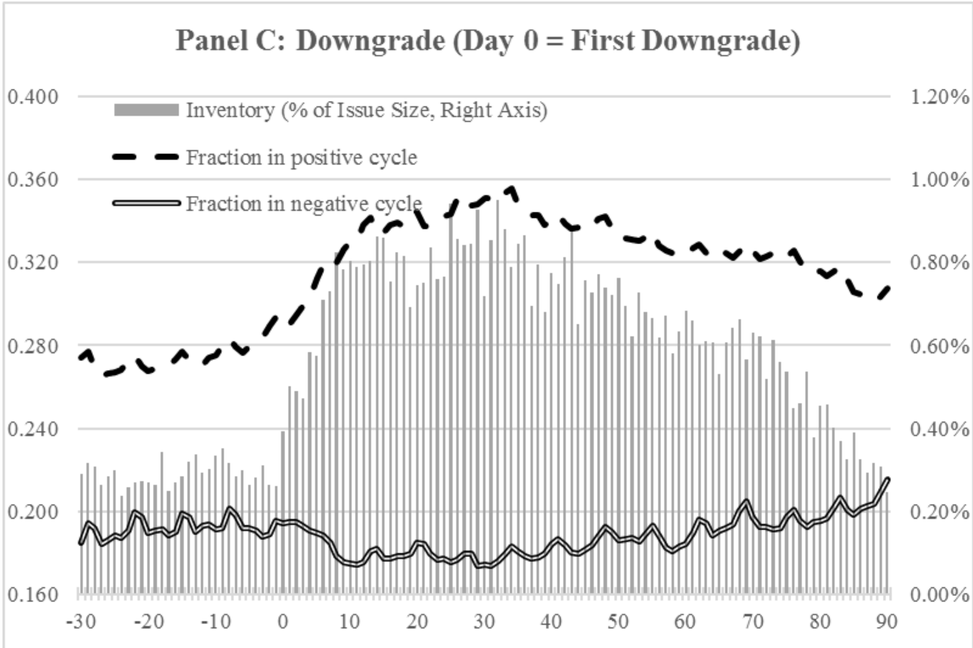


Figure 2. Dealer inventory and cumulative abnormal returns during inventory cycles and around important events. This figure presents average dealer inventory, as a percentage of bond's issue size, and mean/median cumulative abnormal returns, normalized to zero on the event day, during positive (Panel A) and negative (Panel B) inventory cycles, around the downgrades of corporate bonds from investment to speculative grades (Panel C), and in the periods in which the accumulative abnormal returns for the bonds are highly negative (Panel D). In Panels A and B, the event day (day 0) is the day on which the inventory reaches the cycle peak. In Panel C, the event day is the day on which the bond is first downgraded from investment to speculative grades by at least one of the three rating agencies-- S&P, Moody's, and Fitch. In Panel D, the event day is the first day on which the three-month cumulative abnormal returns of the bonds reach the levels that put them in the bottom decile of the sample.



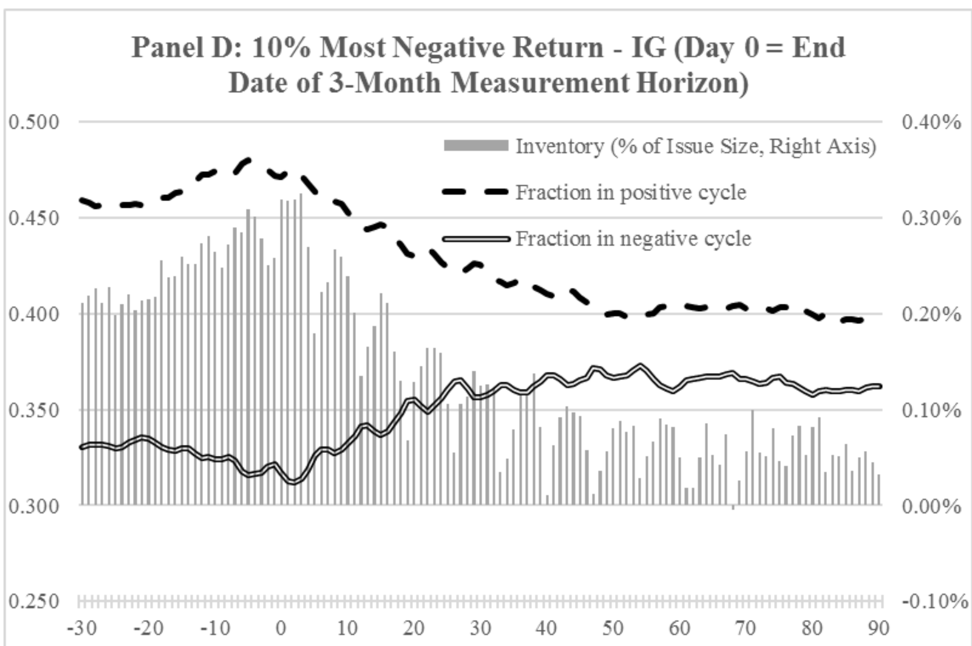


Figure 2 *-continued.*

In all panels, only investment-grade bonds with maturity of at least one year and age of at least one year on the event day are included. Except in Panel C (due to relatively few events), an additional inclusion requirement that the bond must have issue size of at least \$1 billion is also imposed. Bond return is calculated as percentage change in VWAP between two trading days, winsorized at 2.5% in both tails, and abnormal return is obtained by subtracting the bond return by Bank of America Merrill Lynch's Investment Grade Corporate Bond Index return.